Neural Computation -- 2009

136 article(s), all issues

1. Nonnegative matrix factorization with the itakura-saito divergence

FévotteCédric, BertinNancy, DurrieuJean-Louis

https://doi.org/10.5555/2696354.2696362

This letter presents theoretical, algorithmic, and experimental results about nonnegative matrix factorization (NMF) with the Itakura-Saito (IS) divergence. We describe how IS-NMF is underlaid by a...

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2. Capacity of a single spiking neuron

Shiro Ikeda, Jonathan H. Manton

(no DOI)

Information transfer through a single neuron is a fundamental component of information processing in the brain, and computing the information channel capacity is important to understand this information processing. The problem is difficult since the capacity depends on coding, characteristics of the communication channel, and optimization over input distributions, among other issues. In this letter, we consider two models. The temporal coding model of a neuron as a communication channel assumes the output is tau where tau is a gamma-distributed random variable corresponding to the interspike interval, that is, the time it takes for the neuron to fire once. The rate coding model is similar; the output is the actual rate of firing over a fixed period of time. Theoretical studies prove that the distribution of inputs, which achieves channel capacity, is a discrete distribution with finite mass points for temporal and rate coding under a reasonable assumption. This allows us to compute numerically the capacity of a neuron. Numerical results are in a plausible range based on biological evidence to date.

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3. Long-range out-of-sample properties of autoregressive neural networks

LeoniPatrick

https://doi.org/10.5555/2696343.2696344

We consider already-trained discrete autoregressive neural networks in their most general representations, with the exclusion of time-varying input though, and we provide tight sufficient condition...

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4. Complete stability in delayed neural networks

■■■*, Chih-Wen Shih*

(no DOI)

No abstract available.

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5. How to Modify a Neural Network Gradually Without Changing Its Input-Output Functionality

Christopher DiMattina, Kechen Zhang

https://doi.org/10.1162/neco.2009.05-08-781 | pp. 1-47

It is generally unknown when distinct neural networks having different synaptic weights and thresholds implement identical input-output transformations. Determining the exact conditions for structurally distinct yet functionally equivalent networks may shed light on the theoretical constraints on how diverse neural circuits might develop and be maintained to serve identical functions. Such consideration also imposes practical limits on our ability to uniquely infer the structure of underlying neural circuits from stimulus-response measurements. We introduce a biologically inspired mathematical method for determining when the structure of a neural network can be perturbed gradually while preserving functionality. We show that for common three-layer networks with convergent and nondegenerate connection weights, this is possible only when the hidden unit gains are power functions, exponentials, or logarithmic functions, which are known to approximate the gains seen in some biological neurons. For practical applications, our numerical simulations with finite and noisy data show that continuous confounding of parameters due to network functional equivalence tends to occur approximately even when the gain function is not one of the aforementioned three types, suggesting that our analytical results are applicable to more general situations and may help identify a common source of parameter variability in neural network modeling.

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6. Current Source Density Reconstruction from Incomplete Data

Daniel K. Wójcik, Szymon ■■ski

https://doi.org/10.1162/neco.2009.07-08-831 | pp. 48-60

We propose two ways of estimating current source density (CSD) from measurements of voltage on a Cartesian grid with missing recording points using the inverse CSD method. The simplest approach is to substitute local averages (LA) in place of missing data. A more elaborate alternative is to estimate a smaller number of CSD parameters than the actual number of recordings and to take the least-squares fit (LS). We compare the two approaches in the three-dimensional case on several sets of surrogate and experimental data, for varying numbers of missing data points, and discuss their advantages and drawbacks. One can construct CSD distributions for which one or the other approach is better. However, in general, the LA method is to be recommended as being more stable and more robust to variations in the recorded fields.

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7. Spiking Neuron Model for Temporal Sequence Recognition

Sean Byrnes, Anthony N. Burkitt, David B. Grayden, Hamish Meffin

https://doi.org/10.1162/neco.2009.12-07-679 | pp. 61-93

A biologically inspired neuronal network that stores and recognizes temporal sequences of symbols is described. Each symbol is represented by excitatory input to distinct groups of neurons (symbol pools). Unambiguous storage of multiple sequences with common subsequences is ensured by partitioning each symbol pool into subpools that respond only when the current symbol has been preceded by a particular sequence of symbols. We describe synaptic structure and neural dynamics that permit the selective activation of subpools by the correct sequence. Symbols may have varying durations of the order of

hundreds of milliseconds. Physiologically plausible plasticity mechanisms operate on a time scale of tens of milliseconds; an interaction of the excitatory input with periodic global inhibition bridges this gap so that neural events representing successive symbols occur on this much faster timescale. The network is shown to store multiple overlapping sequences of events. It is robust to variation in symbol duration, it is scalable, and its performance degrades gracefully with perturbation of its parameters.

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8. Mean, Variance, and Autocorrelation of Subthreshold Potential Fluctuations Driven by Filtered Conductance Shot Noise

Lars Wolff, Benjamin Lindner

https://doi.org/10.1162/neco.2009.02-09-958 | pp. 94-120

We study the subthreshold voltage fluctuations of a conductance-based passive point neuron stimulated by filtered Poissonian shot noise. We give exact analytical expressions in terms of quadratures for the first two time-dependent moments and the autocorrelation function of the membrane voltage. We also derive simplified expressions for the moments in terms of elementary functions that hold true in the limit case of short filter time, small spike amplitude, and a single synaptic reversal potential. By means of these expressions, we show that for an ensemble of equilibrated conductances but sharp initial voltage (corresponding to a short voltage clamp at the initial time), the mean and the standard deviation can display nonmonotonic time courses. In particular, transient changes in the standard deviation disagree strongly with the predictions of the commonly used effective time constant approximation over a large parameter range. We also study the dependence of the correlation time of the voltage on the synaptic spike amplitude and the synaptic input rate. All results are confirmed by extensive stochastic simulations.

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9. The Computational Structure of Spike Trains

Robert Haslinger, Kristina Lisa Klinkner, Cosma Rohilla Shalizi

https://doi.org/10.1162/neco.2009.12-07-678 | pp. 121-157

Neurons perform computations, and convey the results of those computations through the statistical structure of their output spike trains. Here we present a practical method, grounded in the information-theoretic analysis of prediction, for inferring a minimal representation of that structure and for characterizing its complexity. Starting from spike trains, our approach finds their causal state models (CSMs), the minimal hidden Markov models or stochastic automata capable of generating statistically identical time series. We then use these CSMs to objectively quantify both the generalizable structure and the idiosyncratic randomness of the spike train. Specifically, we show that the expected algorithmic information content (the information needed to describe the spike train exactly) can be split into three parts describing (1) the time-invariant structure (complexity) of the minimal spike-generating process, which describes the spike train statistically; (2) the randomness (internal entropy rate) of the minimal spike-generating process; and (3) a residual pure noise term not described by the minimal spike-generating process. We use CSMs to approximate each of these quantities. The CSMs are inferred nonparametrically from the data, making only mild regularity assumptions, via the causal state splitting reconstruction algorithm. The methods presented here complement more traditional spike train analyses by describing not only spiking probability and spike train entropy, but also the complexity of a spike train's structure. We demonstrate our approach using both simulated spike trains and experimental data recorded in rat barrel

cortex during vibrissa stimulation.

10. On the Use of Dynamic Bayesian Networks in Reconstructing Functional Neuronal Networks from Spike Train Ensembles

Seif Eldawlatly, Yang Zhou, Rong Jin, Karim Oweiss
https://doi.org/10.1162/neco.2009.11-08-900 | pp. 158-189

Coordination among cortical neurons is believed to be a key element in mediating many high-level cortical processes such as perception, attention, learning, and memory formation. Inferring the structure of the neural circuitry underlying this coordination is important to characterize the highly nonlinear, time-varying interactions between cortical neurons in the presence of complex stimuli. In this work, we investigate the applicability of dynamic Bayesian networks (DBNs) in inferring the effective connectivity between spiking cortical neurons from their observed spike trains. We demonstrate that DBNs can infer the underlying nonlinear and time-varying causal interactions between these neurons and can discriminate between mono- and polysynaptic links between them under certain constraints governing their putative connectivity. We analyzed conditionally Poisson spike train data mimicking spiking activity of cortical networks of small and moderately large size. The performance was assessed and compared to other methods under systematic variations of the network structure to mimic a wide range of responses typically observed in the cortex. Results demonstrate the utility of DBN in inferring the effective connectivity in cortical networks.

11. Visuotactile Representation of Peripersonal Space: A Neural Network Study

Elisa Magosso, Melissa Zavaglia, Andrea Serino, Giuseppe di Pellegrino, Mauro Ursino
https://doi.org/10.1162/neco.2009.01-08-694 | pp. 190-243

Neurophysiological and behavioral studies suggest that the peripersonal space is represented in a multisensory fashion by integrating stimuli of different modalities. We developed a neural network to simulate the visual-tactile representation of the peripersonal space around the right and left hands. The model is composed of two networks (one per hemisphere), each with three areas of neurons: two are unimodal (visual and tactile) and communicate by synaptic connections with a third downstream multimodal (visual-tactile) area. The hemispheres are interconnected by inhibitory synapses. We applied a combination of analytic and computer simulation techniques. The analytic approach requires some simplifying assumptions and approximations (linearization and a reduced number of neurons) and is used to investigate network stability as a function of parameter values, providing some emergent properties. These are then tested and extended by computer simulations of a more complex nonlinear network that does not rely on the previous simplifications. With basal parameter values, the extended network reproduces several in vivo phenomena: multisensory coding of peripersonal space, reinforcement of unisensory perception by multimodal stimulation, and coexistence of simultaneous right- and left-hand representations in bilateral stimulation. By reducing the strength of the synapses from the right tactile neurons, the network is able to mimic the responses characteristic of right-brain-damaged patients with left tactile extinction: perception of unilateral left tactile stimulation, cross-modal extinction and cross-modal facilitation in bilateral stimulation. Finally, a variety of sensitivity analyses on some key parameters was performed to shed light on the contribution of single-model components in network behaviour. The model may help us understand the

neural circuitry underlying peripersonal space representation and identify its alterations explaining neurological deficits. In perspective, it could help in interpreting results of psychophysical and behavioral trials and clarifying the neural correlates of multisensory-based rehabilitation procedures.

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12. Discrete States of Synaptic Strength in a Stochastic Model of Spike-Timing-Dependent Plasticity

Terry Elliott

https://doi.org/10.1162/neco.2009.07-08-814 | pp. 244-272

A stochastic model of spike-timing-dependent plasticity (STDP) postulates that single synapses presented with a single spike pair exhibit all-or-none quantal jumps in synaptic strength. The amplitudes of the jumps are independent of spiking timing, but their probabilities do depend on spiking timing. By making the amplitudes of both upward and downward transitions equal, synapses then occupy only a discrete set of states of synaptic strength. We explore the impact of a finite, discrete set of strength states on our model, finding three principal results. First, a finite set of strength states limits the capacity of a single synapse to express the standard, exponential STDP curve. We derive the expression for the expected change in synaptic strength in response to a standard, experimental spike pair protocol, finding a deviation from exponential behavior. We fit our prediction to recent data from single dendritic spine heads, finding results that are somewhat better than exponential fits. Second, we show that the fixed-point dynamics of our model regulate the upward and downward transition probabilities so that these are on average equal, leading to a uniform distribution of synaptic strength states. However, third, under long-term potentiation (LTP) and long-term depression (LTD) protocols, these probabilities are unequal, skewing the distribution away from uniformity. If the number of states of strength is at least of order 10, then we find that three effective states of synaptic strength appear, consistent with some experimental data on ternary-strength synapses. On this view, LTP and LTD protocols may therefore be saturating protocols.

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13. Hebbian Self-Organizing Integrate-and-Fire Networks for Data Clustering

Florian Landis, Thomas Ott, Ruedi Stoop

https://doi.org/10.1162/neco.2009.12-08-926 | pp. 273-288

We propose a Hebbian learning-based data clustering algorithm using spiking neurons. The algorithm is capable of distinguishing between clusters and noisy background data and finds an arbitrary number of clusters of arbitrary shape. These properties render the approach particularly useful for visual scene segmentation into arbitrarily shaped homogeneous regions. We present several application examples, and in order to highlight the advantages and the weaknesses of our method, we systematically compare the results with those from standard methods such as the k-means and Ward's linkage clustering. The analysis demonstrates that not only the clustering ability of the proposed algorithm is more powerful than those of the two concurrent methods, the time complexity of the method is also more modest than that of its generally used strongest competitor.

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14. Memory Capacities for Synaptic and Structural Plasticity

Andreas Knoblauch, Günther Palm, Friedrich T. Sommer

https://doi.org/10.1162/neco.2009.08-07-588 | pp. 289-341

Neural associative networks with plastic synapses have been proposed as computational models of brain functions and also for applications such as pattern recognition and information retrieval. To guide biological models and optimize technical applications, several definitions of memory capacity have been used to measure the efficiency of associative memory. Here we explain why the currently used performance measures bias the comparison between models and cannot serve as a theoretical benchmark. We introduce fair measures for information-theoretic capacity in associative memory that also provide a theoretical benchmark. In neural networks, two types of manipulating synapses can be discerned: synaptic plasticity, the change in strength of existing synapses, and structural plasticity, the creation and pruning of synapses. One of the new types of memory capacity we introduce permits quantifying how structural plasticity can increase the network efficiency by compressing the network structure, for example, by pruning unused synapses. Specifically, we analyze operating regimes in the Willshaw model in which structural plasticity can compress the network structure and push performance to the theoretical benchmark. The amount C of information stored in each synapse can scale with the logarithm of the network size rather than being constant, as in classical Willshaw and Hopfield nets (< or = ln 2 approximately 0.7). Further, the review contains novel technical material: a capacity analysis of the Willshaw model that rigorously controls for the level of retrieval quality, an analysis for memories with a nonconstant number of active units (where C < or = 1/e ln 2 approximately 0.53), and the analysis of the computational complexity of associative memories with and without network compression.

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15. Derivatives of Logarithmic Stationary Distributions for Policy Gradient Reinforcement Learning

Tetsuro Morimura, Eiji Uchibe, Junichiro Yoshimoto, Jan Peters, Kenji Doya

https://doi.org/10.1162/neco.2009.12-08-922 | pp. 342-376

Most conventional policy gradient reinforcement learning (PGRL) algorithms neglect (or do not explicitly make use of) a term in the average reward gradient with respect to the policy parameter. That term involves the derivative of the stationary state distribution that corresponds to the sensitivity of its distribution to changes in the policy parameter. Although the bias introduced by this omission can be reduced by setting the forgetting rate gamma for the value functions close to 1, these algorithms do not permit gamma to be set exactly at gamma = 1. In this article, we propose a method for estimating the log stationary state distribution derivative (LSD) as a useful form of the derivative of the stationary state distribution through backward Markov chain formulation and a temporal difference learning framework. A new policy gradient (PG) framework with an LSD is also proposed, in which the average reward gradient can be estimated by setting gamma = 0, so it becomes unnecessary to learn the value functions. We also test the performance of the proposed algorithms using simple benchmark tasks and show that these can improve the performances of existing PG methods.

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16. Systematic Fluctuation Expansion for Neural Network Activity Equations

Michael A. Buice, Jack D. Cowan, Carson C. Chow

https://doi.org/10.1162/neco.2009.02-09-960 | pp. 377-426

Population rate or activity equations are the foundation of a common approach to modeling for neural networks. These equations provide mean field dynamics for the firing rate or activity of neurons within a network given some connectivity. The shortcoming of these equations is that they take into account only the average firing rate, while leaving out higher-order statistics like correlations between firing. A stochastic theory of neural networks that includes statistics at all orders was recently formulated. We describe how this theory yields a systematic extension to population rate equations by introducing equations for correlations and appropriate coupling terms. Each level of the approximation yields closed equations; they depend only on the mean and specific correlations of interest, without an ad hoc criterion for doing so. We show in an example of an all-to-all connected network how our system of generalized activity equations captures phenomena missed by the mean field rate equations alone.

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17. Cross-Correlations in High-Conductance States of a Model Cortical Network

John Hertz

https://doi.org/10.1162/neco.2009.06-08-806 | pp. 427-447

Neuronal firing correlations are studied using simulations of a simple network model for a cortical column in a high-conductance state with dynamically balanced excitation and inhibition. Although correlations between individual pairs of neurons exhibit considerable heterogeneity, population averages show systematic behavior. When the network is in a stationary state, the average correlations are generically small: correlation coefficients are of order 1/N, where N is the number of neurons in the network. However, when the input to the network varies strongly in time, much larger values are found. In this situation, the network is out of balance, and the synaptic conductance is low, at times when the strongest firing occurs. However, examination of the correlation functions of synaptic currents reveals that after these bursts, balance is restored within a few milliseconds by a rapid increase in inhibitory synaptic conductance. These findings suggest an extension of the notion of the balanced state to include balanced fluctuations of synaptic currents, with a characteristic timescale of a few milliseconds.

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18. Dynamics and Robustness of Familiarity Memory

Jesús M. Cortés, Andrea Greve, Adam B. Barrett, Mark C. W. van Rossum

https://doi.org/10.1162/neco.2009.12-08-921 | pp. 448-466

When presented with an item or a face, one might have a sense of recognition without the ability to recall when or where the stimulus has been encountered before. This sense of recognition is called familiarity memory. Following previous computational studies of familiarity memory, we investigate the dynamical properties of familiarity discrimination and contrast two different familiarity discriminators: one based on the energy of the neural network and the other based on the time derivative of the energy. We show how the familiarity signal decays rapidly after stimulus presentation. For both discriminators, we calculate the capacity using mean field analysis. Compared to recall capacity (the classical associative memory in Hopfield nets), both the energy and the slope discriminators have bigger capacity, yet the energy-based discriminator has a higher capacity than one based on its time derivative. Finally, both discriminators are found to have a different noise dependence.

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19. Supervised Learning in Spiking Neural Networks with ReSuMe: Sequence Learning, Classification, and Spike Shifting

Filip Ponulak, Andrzej Kasi■ski

https://doi.org/10.1162/neco.2009.11-08-901 | pp. 467-510

Learning from instructions or demonstrations is a fundamental property of our brain necessary to acquire new knowledge and develop novel skills or behavioral patterns. This type of learning is thought to be involved in most of our daily routines. Although the concept of instruction-based learning has been studied for several decades, the exact neural mechanisms implementing this process remain unrevealed. One of the central questions in this regard is, How do neurons learn to reproduce template signals (instructions) encoded in precisely timed sequences of spikes? Here we present a model of supervised learning for biologically plausible neurons that addresses this question. In a set of experiments, we demonstrate that our approach enables us to train spiking neurons to reproduce arbitrary template spike patterns in response to given synaptic stimuli even in the presence of various sources of noise. We show that the learning rule can also be used for decision-making tasks. Neurons can be trained to classify categories of input signals based on only a temporal configuration of spikes. The decision is communicated by emitting precisely timed spike trains associated with given input categories. Trained neurons can perform the classification task correctly even if stimuli and corresponding decision times are temporally separated and the relevant information is consequently highly overlapped by the ongoing neural activity. Finally, we demonstrate that neurons can be trained to reproduce sequences of spikes with a controllable time shift with respect to target templates. A reproduced signal can follow or even precede the targets. This surprising result points out that spiking neurons can potentially be applied to forecast the behavior (firing times) of other reference neurons or networks.

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20. Convolutional Networks Can Learn to Generate Affinity Graphs for Image Segmentation

Srinivas C. Turaga, Joseph F. Murray, Viren Jain, Fabian C. Roth, Moritz Helmstaedter, Kevin L. Briggman, Winfried Denk, H. Sebastian Seung

https://doi.org/10.1162/neco.2009.10-08-881 | pp. 511-538

Many image segmentation algorithms first generate an affinity graph and then partition it. We present a machine learning approach to computing an affinity graph using a convolutional network (CN) trained using ground truth provided by human experts. The CN affinity graph can be paired with any standard partitioning algorithm and improves segmentation accuracy significantly compared to standard hand-designed affinity functions. We apply our algorithm to the challenging 3D segmentation problem of reconstructing neuronal processes from volumetric electron microscopy (EM) and show that we are able to learn a good affinity graph directly from the raw EM images. Further, we show that our affinity graph improves the segmentation accuracy of both simple and sophisticated graph partitioning algorithms. In contrast to previous work, we do not rely on prior knowledge in the form of hand-designed image features or image preprocessing. Thus, we expect our algorithm to generalize effectively to arbitrary image types.

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21. On the Effects of Signal Acuity in a Multi-Alternative Model of Decision Making

Tyler McMillen, Sam Behseta

https://doi.org/10.1162/neco.2009.01-09-938 | pp. 539-580

We consider the effects of signal sharpness or acuity on the performance of neural models of decision making. In these models, a vector of signals is presented, and the subject must decide which of the elements of the vector is the largest. McMillen and Holmes (2006) derived asymptotically optimal tests under the assumption that the elements of the signal vector were all equal except one. In this letter, we consider the case of signals spread around a peak. The acuity is a measure of how strongly peaked the signal is. We find that the optimal test is one in which the detectors are passed through an output layer that encodes knowledge of the possible shapes of the incoming signals. The incorporation of such an output layer can lead to significant improvements in decision-making tasks.

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22. Feature Selection in Simple Neurons: How Coding Depends on Spiking Dynamics

Michael Famulare, Adrienne Fairhall

https://doi.org/10.1162/neco.2009.02-09-956 | pp. 581-598

The relationship between a neuron's complex inputs and its spiking output defines the neuron's coding strategy. This is frequently and effectively modeled phenomenologically by one or more linear filters that extract the components of the stimulus that are relevant for triggering spikes and a nonlinear function that relates stimulus to firing probability. In many sensory systems, these two components of the coding strategy are found to adapt to changes in the statistics of the inputs in such a way as to improve information transmission. Here, we show for two simple neuron models how feature selectivity as captured by the spike-triggered average depends on both the parameters of the model and the statistical characteristics of the input.

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23. Phase Locking Below Rate Threshold in Noisy Model Neurons

Jan A. Freund, Alexander P. Nikitin, Nigel G. Stocks

https://doi.org/10.1162/neco.2009.01-09-934 | pp. 599-620

The property of a neuron to phase-lock to an oscillatory stimulus before adapting its spike rate to the stimulus frequency plays an important role for the auditory system. We investigate under which conditions neurons exhibit this phase locking below rate threshold. To this end, we simulate neurons employing the widely used leaky integrate-and-fire (LIF) model. Tuning parameters, we can arrange either an irregular spontaneous or a tonic spiking mode. When the neuron is stimulated in both modes, a significant rise of vector strength prior to a noticeable change of the spike rate can be observed. Combining analytic reasoning with numerical simulations, we trace this observation back to a modulation of interspike intervals, which itself requires spikes to be only loosely coupled. We test the limits of this conception by simulating an LIF model with threshold fatigue, which generates pronounced anticorrelations between subsequent interspike intervals. In addition we evaluate the LIF response for harmonic stimuli of various frequencies and discuss the extension to more complex stimuli. It seems that phase locking below rate threshold occurs generically for all zero mean stimuli. Finally, we discuss our findings in the context of stimulus detection.

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24. Population Models of Temporal Differentiation

Bryan Tripp, Chris Eliasmith

https://doi.org/10.1162/neco.2009.02-09-970 | pp. 621-659

Temporal derivatives are computed by a wide variety of neural circuits, but the problem of performing this computation accurately has received little theoretical study. Here we systematically compare the performance of diverse networks that calculate derivatives using cell-intrinsic adaptation and synaptic depression dynamics, feedforward network dynamics, and recurrent network dynamics. Examples of each type of network are compared by quantifying the errors they introduce into the calculation and their rejection of high-frequency input noise. This comparison is based on both analytical methods and numerical simulations with spiking leaky-integrate-and-fire (LIF) neurons. Both adapting and feedforward-network circuits provide good performance for signals with frequency bands that are well matched to the time constants of postsynaptic current decay and adaptation, respectively. The synaptic depression circuit performs similarly to the adaptation circuit, although strictly speaking, precisely linear differentiation based on synaptic depression is not possible, because depression scales synaptic weights multiplicatively. Feedback circuits introduce greater errors than functionally equivalent feedforward circuits, but they have the useful property that their dynamics are determined by feedback strength. For this reason, these circuits are better suited for calculating the derivatives of signals that evolve on timescales outside the range of membrane dynamics and, possibly, for providing the wide range of timescales needed for precise fractional-order differentiation.

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25. Precise Capacity Analysis in Binary Networks with Multiple Coding Level Inputs

Yali Amit, Yibi Huang

https://doi.org/10.1162/neco.2009.02-09-967 | pp. 660-688

We compute retrieval probabilities as a function of pattern age for networks with binary neurons and synapses updated with the simple Hebbian learning model studied in Amit and Fusi (1994). The analysis depends on choosing a neural threshold that enables patterns to stabilize in the neural dynamics. In contrast to most earlier work, where selective neurons for each pattern are drawn independently with fixed probability f, here we analyze the situation where f is drawn from some distribution on a range of coding levels. In order to set a workable threshold in this setting, it is necessary to introduce a simple inhibition in the neural dynamics whose magnitude depends on the total activity of the network. Proper choice of the threshold depends on the value of the covariances between the synapses for which we provide an explicit formula. Retrieval probabilities depend on the distribution of the fields induced by a learned pattern. We show that the field induced by the first learned pattern evolves as a Markov chain during subsequent learning epochs, leading to a recursive formula for the distribution. Alternatively, the distribution can be computed using a normal approximation, which involves the value of the synaptic covariances. Capacity is computed as the sum of the retrieval probabilities over all ages. We show through simulation that the chosen threshold enables retrieval with asynchronous dynamics even in the presence of significant noise in the initial state of the pattern. The computed probabilities with both methods are shown to be very close to probabilities estimated from simulation. The analysis is extended to randomly connected networks.

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26. Growing Self-Organizing Surface Map: Learning a Surface Topology from a Point Cloud

Vilson Luiz Dalle Mole, A.F.R. Araujo

https://doi.org/10.1162/neco.2009.08-08-842 | pp. 689-729

The growing self-organizing surface map (GSOSM) is a novel map model that learns a folded surface immersed in a 3D space. Starting from a dense point cloud, the surface is reconstructed through an incremental mesh composed of approximately equilateral triangles. Unlike other models such as neural meshes (NM), the GSOSM builds a surface topology while accepting any sequence of sample presentation. The GSOSM model introduces a novel connection learning rule called competitive connection Hebbian learning (CCHL), which produces a complete triangulation. GSOSM reconstructions are accurate and often free of false or overlapping faces. This letter presents and discusses the GSOSM model. It also presents and analyzes a set of results and compares GSOSM with some other models.

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27. Autonomous Development of Vergence Control Driven by Disparity Energy Neuron Populations

Yiwen Wang, Bertram E. Shi

https://doi.org/10.1162/neco.2009.01-09-950 | pp. 730-751

We present a simple optimization criterion that leads to autonomous development of a sensorimotor feedback loop driven by the neural representation of the depth in the mammalian visual cortex. Our test bed is an active stereo vision system where the vergence angle between the two eyes is controlled by the output of a population of disparity-selective neurons. By finding a policy that maximizes the total response across the neuron population, the system eventually tracks a target as it moves in depth. We characterized the tracking performance of the resulting policy using objects moving both sinusoidally and randomly in depth. Surprisingly, the system can even learn how to track based on stimuli it cannot track: even though the closed loop 3 dB tracking bandwidth of the system is 0.3 Hz, correct tracking policies are learned for input stimuli moving as fast as 0.75 Hz.

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28. A Moving Bump in a Continuous Manifold: A Comprehensive Study of the Tracking Dynamics of Continuous Attractor Neural Networks

C. C. Alan Fung, K. Y. Michael Wong, Si Wu

https://doi.org/10.1162/neco.2009.07-08-824 | pp. 752-792

Understanding how the dynamics of a neural network is shaped by the network structure and, consequently, how the network structure facilitates the functions implemented by the neural system is at the core of using mathematical models to elucidate brain functions. This study investigates the tracking dynamics of continuous attractor neural networks (CANNs). Due to the translational invariance of neuronal recurrent interactions, CANNs can hold a continuous family of stationary states. They form a continuous manifold in which the neural system is neutrally stable. We systematically explore how this property facilitates the tracking performance of a CANN, which is believed to have clear correspondence with brain functions. By using the wave functions of the quantum harmonic oscillator as the basis, we demonstrate how the dynamics of a CANN is decomposed into different motion modes, corresponding to distortions in the

amplitude, position, width, or skewness of the network state. We then develop a perturbation approach that utilizes the dominating movement of the network's stationary states in the state space. This method allows us to approximate the network dynamics up to an arbitrary accuracy depending on the order of perturbation used. We quantify the distortions of a gaussian bump during tracking and study their effects on tracking performance. Results are obtained on the maximum speed for a moving stimulus to be trackable and the reaction time for the network to catch up with an abrupt change in the stimulus.

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29. Regularization Techniques and Suboptimal Solutions to Optimization Problems in Learning from Data

Giorgio Gnecco, Marcello Sanguineti

https://doi.org/10.1162/neco.2009.05-08-786 | pp. 793-829

Various regularization techniques are investigated in supervised learning from data. Theoretical features of the associated optimization problems are studied, and sparse suboptimal solutions are searched for. Rates of approximate optimization are estimated for sequences of suboptimal solutions formed by linear combinations of n-tuples of computational units, and statistical learning bounds are derived. As hypothesis sets, reproducing kernel Hilbert spaces and their subsets are considered.

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30. Efficient Learning and Feature Selection in High-Dimensional Regression

Jo-Anne Ting, Aaron D'Souza, Sethu Vijayakumar, Stefan Schaal

https://doi.org/10.1162/neco.2009.02-08-702 | pp. 831-886

We present a novel algorithm for efficient learning and feature selection in high-dimensional regression problems. We arrive at this model through a modification of the standard regression model, enabling us to derive a probabilistic version of the well-known statistical regression technique of backfitting. Using the expectation-maximization algorithm, along with variational approximation methods to overcome intractability, we extend our algorithm to include automatic relevance detection of the input features. This variational Bayesian least squares (VBLS) approach retains its simplicity as a linear model, but offers a novel statistically robust black-box approach to generalized linear regression with high-dimensional inputs. It can be easily extended to nonlinear regression and classification problems. In particular, we derive the framework of sparse Bayesian learning, the relevance vector machine, with VBLS at its core, offering significant computational and robustness advantages for this class of methods. The iterative nature of VBLS makes it most suitable for real-time incremental learning, which is crucial especially in the application domain of robotics, brain-machine interfaces, and neural prosthetics, where real-time learning of models for control is needed. We evaluate our algorithm on synthetic and neurophysiological data sets, as well as on standard regression and classification benchmark data sets, comparing it with other competitive statistical approaches and demonstrating its suitability as a drop-in replacement for other generalized linear regression techniques.

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31. Adaptive Design Optimization: A Mutual Information-Based Approach to Model Discrimination in Cognitive Science

Daniel R. Cavagnaro, Jay I. Myung, Mark A. Pitt, Janne V. Kujala
https://doi.org/10.1162/neco.2009.02-09-959 | pp. 887-905

Discriminating among competing statistical models is a pressing issue for many experimentalists in the field of cognitive science. Resolving this issue begins with designing maximally informative experiments. To this end, the problem to be solved in adaptive design optimization is identifying experimental designs under which one can infer the underlying model in the fewest possible steps. When the models under consideration are nonlinear, as is often the case in cognitive science, this problem can be impossible to solve analytically without simplifying assumptions. However, as we show in this letter, a full solution can be found numerically with the help of a Bayesian computational trick derived from the statistics literature, which recasts the problem as a probability density simulation in which the optimal design is the mode of the density. We use a utility function based on mutual information and give three intuitive interpretations of the utility function in terms of Bayesian posterior estimates. As a proof of concept, we offer a simple example application to an experiment on memory retention.

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32. EEG-fMRI Fusion of Paradigm-Free Activity Using Kalman Filtering

Thomas Deneux, Olivier Faugeras
https://doi.org/10.1162/neco.2009.05-08-793 | pp. 906-948

We address here the use of EEG and fMRI, and their combination, in order to estimate the full spatiotemporal patterns of activity on the cortical surface in the absence of any particular assumptions on this activity such as stimulation times. For handling such a high-dimension inverse problem, we propose the use of (1) a global forward model of how these measures are functions of the "neural activity" of a large number of sources distributed on the cortical surface, formalized as a dynamical system, and (2) adaptive filters, as a natural solution to solve this inverse problem iteratively along the temporal dimension. This estimation framework relies on realistic physiological models, uses EEG and fMRI in a symmetric manner, and takes into account both their temporal and spatial information. We use the Kalman filter and smoother to perform such an estimation on realistic artificial data and demonstrate that the algorithm can handle the high dimensionality of these data and that it succeeds in solving this inverse problem, combining efficiently the information provided by the two modalities (this information being naturally predominantly temporal for EEG and spatial for fMRI). It performs particularly well in reconstructing a random temporally and spatially smooth activity spread over the cortex. The Kalman filter and smoother show some limitations, however, which call for the development of more specific adaptive filters. First, they do not cope well with the strong nonlinearity in the model that is necessary for an adequate description of the relation between cortical electric activities and the metabolic demand responsible for fMRI signals. Second, they fail to estimate a sparse activity (i.e., presenting sharp peaks at specific locations and times). Finally their computational cost remains high. We use schematic examples to explain these limitations and propose further developments of our method to overcome them.

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33. A Model of Indispensability of a Large Glial Layer in Cerebrovascular Circulation

Rohit Gandrakota, V. Srinivasa Chakravarthy, Ranjan K. Pradhan
https://doi.org/10.1162/neco.2009.01-09-945 | pp. 949-968

We formulate the problem of oxygen delivery to neural tissue as a problem of association. Input to a pool of neurons in one brain area must be matched in space and time with metabolic inputs from the vascular network via the glial network. We thus have a model in which neural, glial, and vascular layers are connected bidirectionally, in that order. Connections between neuro-glial and glial-vascular stages are trained by an unsupervised learning mechanism such that input to the neural layer is sustained by the precisely patterned delivery of metabolic inputs from the vascular layer via the glial layer. Simulations show that the capacity of such a system to sustain patterns is weak when the glial layer is absent. Capacity is higher when a glial layer is present and increases with the layer size. The proposed formulation of neurovascular interactions raises many intriguing questions about the role of glial cells in cerebral circulation.

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34. Dynamical Mean Field Model of a Neural-Glial Mass

Roberto C. Sotero, Ramón Martínez-Cancino

https://doi.org/10.1162/neco.2009.04-09-1002 | pp. 969-997

Our goal is to model the behavior of an ensemble of interacting neurons and astrocytes (the neural-glial mass). For this, a model describing N tripartite synapses is proposed. Each tripartite synapse consists of presynaptic and postsynaptic nerve terminals, as well as the synaptically associated astrocytic microdomain, and is described by a system of 13 stochastic differential equations. Then, by applying the dynamical mean field approximation (DMA) (Hasegawa, 2003a , 2003b) the system of 13N equations is reduced to 13(13 + 2) = 195 deterministic differential equations for the means and the second-order moments of local and global variables. Simulations are carried out for studying the response of the neural-glial mass to external inputs applied to either the presynaptic terminals or the astrocytes. Three cases were considered: the astrocytes influence only the presynaptic terminal, only the postsynaptic terminal, or both the presynaptic and postsynaptic terminals. As a result, a wide range of responses varying from singles spikes to train of spikes was evoked on presynaptic and postsynaptic terminals. The experimentally observed phenomenon of spontaneous activity in astrocytes was replicated on the neural-glial mass. The model predicts that astrocytes can have a strong and activity-dependent influence on synaptic transmission. Finally, simulations show that the dynamics of astrocytes influences the synchronization ratio between neurons, predicting a peak in the synchronization for specific values of the astrocytes' parameters.

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35. Determining Burst Firing Time Distributions from Multiple Spike Trains

Luis F. Lago-Fernández, Attila Szücs, Pablo Varona

https://doi.org/10.1162/neco.2008.07-07-571 | pp. 973-990

Recent experimental findings have shown the presence of robust and cell-type-specific intraburst firing patterns in bursting neurons. We address the problem of characterizing these patterns under the assumption that the bursts exhibit well-defined firing time distributions. We propose a method for estimating these distributions based on a burst alignment algorithm that minimizes the overlap among the firing time distributions of the different spikes within the burst. This method provides a good approximation to the burst's intrinsic temporal structure as a set of firing time distributions. In addition, the method allows labeling the spikes in any particular burst, establishing a correspondence between each spike and the distribution that best explains it, and identifying missing spikes. Our results on both simulated and experimental data

from the lobster stomatogastric ganglion show that the proposed method provides a reliable characterization of the intraburst firing patterns and avoids the errors derived from missing spikes. This method can also be applied to nonbursting neurons as a general tool for the study and the interpretation of firing time distributions as part of a temporal neural code.

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36. A Continuous Entropy Rate Estimator for Spike Trains Using a K-Means-Based Context Tree

Tiger W. Lin, George N. Reeke

https://doi.org/10.1162/neco.2009.11-08-912 | pp. 998-1024

Entropy rate quantifies the change of information of a stochastic process (Cover & Thomas, 2006). For decades, the temporal dynamics of spike trains generated by neurons has been studied as a stochastic process (Barbieri, Quirk, Frank, Wilson, & Brown, 2001; Brown, Frank, Tang, Quirk, & Wilson, 1998; Kass & Ventura, 2001; Metzner, Koch, Wessel, & Gabbiani, 1998; Zhang, Ginzburg, McNaughton, & Sejnowski, 1998). We propose here to estimate the entropy rate of a spike train from an inhomogeneous hidden Markov model of the spike intervals. The model is constructed by building a context tree structure to lay out the conditional probabilities of various subsequences of the spike train. For each state in the Markov chain, we assume a gamma distribution over the spike intervals, although any appropriate distribution may be employed as circumstances dictate. The entropy and confidence intervals for the entropy are calculated from bootstrapping samples taken from a large raw data sequence. The estimator was first tested on synthetic data generated by multiple-order Markov chains, and it always converged to the theoretical Shannon entropy rate (except in the case of a sixth-order model, where the calculations were terminated before convergence was reached). We also applied the method to experimental data and compare its performance with that of several other methods of entropy estimation.

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37. Conditional Probability-Based Significance Tests for Sequential Patterns in Multineuronal Spike Trains

P. S. Sastry, K. P. Unnikrishnan

https://doi.org/10.1162/neco.2009.12-08-928 | pp. 1025-1059

We consider the problem of detecting statistically significant sequential patterns in multineuronal spike trains. These patterns are characterized by ordered sequences of spikes from different neurons with specific delays between spikes. We have previously proposed a data-mining scheme to efficiently discover such patterns, which occur often enough in the data. Here we propose a method to determine the statistical significance of such repeating patterns. The novelty of our approach is that we use a compound null hypothesis that not only includes models of independent neurons but also models where neurons have weak dependencies. The strength of interaction among the neurons is represented in terms of certain pair-wise conditional probabilities. We specify our null hypothesis by putting an upper bound on all such conditional probabilities. We construct a probabilistic model that captures the counting process and use this to derive a test of significance for rejecting such a compound null hypothesis. The structure of our null hypothesis also allows us to rank-order different significant patterns. We illustrate the effectiveness of our approach using spike trains generated with a simulator.

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38. Spiking Neural Networks for Cortical Neuronal Spike Train Decoding

Huijuan Fang, Yongji Wang, Jiping He

https://doi.org/10.1162/neco.2009.10-08-885 | pp. 1060-1085

Recent investigation of cortical coding and computation indicates that temporal coding is probably a more biologically plausible scheme used by neurons than the rate coding used commonly in most published work. We propose and demonstrate in this letter that spiking neural networks (SNN), consisting of spiking neurons that propagate information by the timing of spikes, are a better alternative to the coding scheme based on spike frequency (histogram) alone. The SNN model analyzes cortical neural spike trains directly without losing temporal information for generating more reliable motor command for cortically controlled prosthetics. In this letter, we compared the temporal pattern classification result from the SNN approach with results generated from firing-rate-based approaches: conventional artificial neural networks, support vector machines, and linear regression. The results show that the SNN algorithm can achieve higher classification accuracy and identify the spiking activity related to movement control earlier than the other methods. Both are desirable characteristics for fast neural information processing and reliable control command pattern recognition for neuroprosthetic applications.

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39. A Simulation Study Investigating the Impact of Dendritic Morphology and Synaptic Topology on Neuronal Firing Patterns

Jen-Yung Chen

https://doi.org/10.1162/neco.2009.11-08-913 | pp. 1086-1111

In the brain, complex information interactions among neurons span several spatial and temporal scales, making it extremely difficult to identify the principles governing neural information processing. In this study, we used computational models to investigate the impact of dendritic morphology and synaptic topology on patterns of neuronal firing. We first constructed Hodgkin-Huxley-type neuron models that possessed dendrites with different morphological features. We then simulated the responses of these neurons to a number of spatiotemporal input patterns. The similarity between neuronal responses to different patterned inputs was effectively evaluated by a novel combination of metric space analysis and multidimensional scaling analyses. The results showed that neurons with different morphological or anatomical features exhibit differences in stimulus-specific temporal encoding and firing reliability. These findings support the idea that in addition to biophysical membrane properties, the dendritic morphology and the synaptic topology of a neuron can play a significant role in neuronal information processing and may directly contribute to various brain functions.

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40. Optimal Decision Making on the Basis of Evidence Represented in Spike Trains

Jiaxiang Zhang, Rafa■ Bogacz

https://doi.org/10.1162/neco.2009.05-09-1025 | pp. 1113-1148

Experimental data indicate that perceptual decision making involves integration of sensory evidence in certain cortical areas. Theoretical studies have proposed that the computation in neural decision circuits approximates statistically optimal decision procedures (e.g., sequential probability ratio test) that maximize

the reward rate in sequential choice tasks. However, these previous studies assumed that the sensory evidence was represented by continuous values from gaussian distributions with the same variance across alternatives. In this article, we make a more realistic assumption that sensory evidence is represented in spike trains described by the Poisson processes, which naturally satisfy the mean-variance relationship observed in sensory neurons. We show that for such a representation, the neural circuits involving cortical integrators and basal ganglia can approximate the optimal decision procedures for two and multiple alternative choice tasks.

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41. Stability of Localized Patterns in Neural Fields

Konstantin Doubrovinski, J. Michael Herrmann

https://doi.org/10.1162/neco.2008.11-06-392 | pp. 1125-1144

We investigate two-dimensional neural fields as a model of the dynamics of macroscopic activations in a cortex-like neural system. While the one-dimensional case was treated comprehensively by Amari 30 years ago, two-dimensional neural fields are much less understood. We derive conditions for the stability for the main classes of localized solutions of the neural field equation and study their behavior beyond parameter-controlled destabilization. We show that a slight modification of the original model yields an equation whose stationary states are guaranteed to satisfy the original problem and numerically demonstrate that it admits localized noncircular solutions. Typically, however, only periodic spatial tessellations emerge on destabilization of rotationally invariant solutions.

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42. Bayesian k-Means as a “Maximization-Expectation” Algorithm

Kenichi Kurihara, Max Welling

https://doi.org/10.1162/neco.2008.12-06-421 | pp. 1145-1172

We introduce a new class of "maximization-expectation" (ME) algorithms where we maximize over hidden variables but marginalize over random parameters. This reverses the roles of expectation and maximization in the classical expectation-maximization algorithm. In the context of clustering, we argue that these hard assignments open the door to very fast implementations based on data structures such as kd-trees and conga lines. The marginalization over parameters ensures that we retain the ability to infer model structure (i.e., number of clusters). As an important example, we discuss a top-down Bayesian k-means algorithm and a bottom-up agglomerative clustering algorithm. In experiments, we compare these algorithms against a number of alternative algorithms that have recently appeared in the literature.

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43. A Non-Markovian Random Walk Underlies a Stochastic Model of Spike-Timing-Dependent Plasticity

Terry Elliott

https://doi.org/10.1162/neco.2009.06-09-1038 | pp. 1180-1230

A stochastic model of spike-timing-dependent plasticity (STDP) proposes that spike timing influences the probability but not the amplitude of synaptic strength change at single synapses. The classic, biphasic STDP profile emerges as a spatial average over many synapses presented with a single spike pair or as a

temporal average over a single synapse presented with many spike pairs. We have previously shown that the model accounts for a variety of experimental data, including spike triplet results, and has a number of desirable theoretical properties, including being entirely self-stabilizing in all regions of parameter space. Our earlier analyses of the model have employed cumbersome spike-to-spike averaging arguments to derive results. Here, we show that the model can be reformulated as a non-Markovian random walk in synaptic strength, the step sizes being fixed as postulated. This change of perspective greatly simplifies earlier calculations by integrating out the proposed switch mechanism by which changes in strength are driven and instead concentrating on the changes in strength themselves. Moreover, this change of viewpoint is generative, facilitating further calculations that would be intractable, if not impossible, with earlier approaches. We prepare the machinery here for these later calculations but also briefly indicate how this machinery may be used by considering two particular applications.

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44. Mean-Field Approximations for Coupled Populations of Generalized Linear Model Spiking Neurons with Markov Refractoriness

Taro Toyoizumi, Kamiar Rahnama Rad, Liam Paninski

https://doi.org/10.1162/neco.2008.04-08-757 | pp. 1203-1243

There has recently been a great deal of interest in inferring network connectivity from the spike trains in populations of neurons. One class of useful models that can be fit easily to spiking data is based on generalized linear point process models from statistics. Once the parameters for these models are fit, the analyst is left with a nonlinear spiking network model with delays, which in general may be very difficult to understand analytically. Here we develop mean-field methods for approximating the stimulus-driven firing rates (in both the time-varying and steady-state cases), auto- and cross-correlations, and stimulus-dependent filtering properties of these networks. These approximations are valid when the contributions of individual network coupling terms are small and, hence, the total input to a neuron is approximately gaussian. These approximations lead to deterministic ordinary differential equations that are much easier to solve and analyze than direct Monte Carlo simulation of the network activity. These approximations also provide an analytical way to evaluate the linear input-output filter of neurons and how the filters are modulated by network interactions and some stimulus feature. Finally, in the case of strong refractory effects, the mean-field approximations in the generalized linear model become inaccurate; therefore, we introduce a model that captures strong refractoriness, retains all of the easy fitting properties of the standard generalized linear model, and leads to much more accurate approximations of mean firing rates and cross-correlations that retain fine temporal behaviors.

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45. Decoding Poisson Spike Trains by Gaussian Filtering

Sidney R. Lehky

https://doi.org/10.1162/neco.2009.07-08-823 | pp. 1245-1271

The temporal waveform of neural activity is commonly estimated by low-pass filtering spike train data through convolution with a gaussian kernel. However, the criteria for selecting the gaussian width sigma are not well understood. Given an ensemble of Poisson spike trains generated by an instantaneous firing rate function lambda(t), the problem was to recover an optimal estimate of lambda(t) by gaussian filtering. We provide equations describing the optimal value of sigma using an error minimization criterion and examine

how the optimal sigma varies within a parameter space defining the statistics of inhomogeneous Poisson spike trains. The process was studied both analytically and through simulations. The rate functions lambda(t) were randomly generated, with the three parameters defining spike statistics being the mean of lambda(t), the variance of lambda(t), and the exponent alpha of the Fourier amplitude spectrum 1/f(alpha) of lambda(t). The value of sigma(opt) followed a power law as a function of the pooled mean interspike interval I, sigma(opt) = aI(b), where a was inversely related to the coefficient of variation C(V) of lambda(t), and b was inversely related to the Fourier spectrum exponent alpha. Besides applications for data analysis, optimal recovery of an analog signal waveform lambda(t) from spike trains may also be useful in understanding neural signal processing in vivo.

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46. Competitive STDP-Based Spike Pattern Learning

Timothée Masquelier, Rudy Guyonneau, Simon J. Thorpe

https://doi.org/10.1162/neco.2008.06-08-804 | pp. 1259-1276

Recently it has been shown that a repeating arbitrary spatiotemporal spike pattern hidden in equally dense distracter spike trains can be robustly detected and learned by a single neuron equipped with spike-timing-dependent plasticity (STDP) (Masquelier, Guyonneau, & Thorpe, 2008). To be precise, the neuron becomes selective to successive coincidences of the pattern. Here we extend this scheme to a more realistic scenario with multiple repeating patterns and multiple STDP neurons "listening" to the incoming spike trains. These "listening" neurons are in competition: as soon as one fires, it strongly inhibits the others through lateral connections (one-winner-take-all mechanism). This tends to prevent the neurons from learning the same (parts of the) repeating patterns, as shown in simulations. Instead, the population self-organizes, trying to cover the different patterns or coding one pattern by the successive firings of several neurons, and a powerful distributed coding scheme emerges. Taken together, these results illustrate how the brain could easily encode and decode information in the spike times, a theory referred to as temporal coding, and how STDP could play a key role by detecting repeating patterns and generating selective response to them.

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47. Connectivity, Dynamics, and Memory in Reservoir Computing with Binary and Analog Neurons

Lars Büsing, Benjamin Schrauwen, Robert Legenstein

https://doi.org/10.1162/neco.2009.01-09-947 | pp. 1272-1311

Reservoir computing (RC) systems are powerful models for online computations on input sequences. They consist of a memoryless readout neuron that is trained on top of a randomly connected recurrent neural network. RC systems are commonly used in two flavors: with analog or binary (spiking) neurons in the recurrent circuits. Previous work indicated a fundamental difference in the behavior of these two implementations of the RC idea. The performance of an RC system built from binary neurons seems to depend strongly on the network connectivity structure. In networks of analog neurons, such clear dependency has not been observed. In this letter, we address this apparent dichotomy by investigating the influence of the network connectivity (parameterized by the neuron in-degree) on a family of network models that interpolates between analog and binary networks. Our analyses are based on a novel estimation of the Lyapunov exponent of the network dynamics with the help of branching process theory, rank measures that

estimate the kernel quality and generalization capabilities of recurrent networks, and a novel mean field predictor for computational performance. These analyses reveal that the phase transition between ordered and chaotic network behavior of binary circuits qualitatively differs from the one in analog circuits, leading to differences in the integration of information over short and long timescales. This explains the decreased computational performance observed in binary circuits that are densely connected. The mean field predictor is also used to bound the memory function of recurrent circuits of binary neurons.

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48. Representation Sharpening Can Explain Perceptual Priming

Samat Moldakarimov, Maxim Bazhenov, Terrence J. Sejnowski

https://doi.org/10.1162/neco.2009.04-09-999 | pp. 1312-1332

Perceiving and identifying an object is improved by prior exposure to the object. This perceptual priming phenomenon is accompanied by reduced neural activity. But whether suppression of neuronal activity with priming is responsible for the improvement in perception is unclear. To address this problem, we developed a rate-based network model of visual processing. In the model, decreased neural activity following priming was due to stimulus-specific sharpening of representations taking place in the early visual areas. Representation sharpening led to decreased interference of representations in higher visual areas that facilitated selection of one of the competing representations, thereby improving recognition. The model explained a wide range of psychophysical and physiological data observed in priming experiments, including antipriming phenomena, and predicted two functionally distinct stages of visual processing.

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49. Increasing Memory Capacity and Reducing Spurious States in Neural Networks by Introducing Coherent and Collective Firing

Yang Wei Koh, Kazuo Takatsuka

https://doi.org/10.1162/neco.2008.07-08-827 | pp. 1321-1334

It is well known that higher-order Hopfield nets called multispin models can increase memory capacity to some extent by extending the direct product of spin states to more than second order. However, a group of neurons can then respond degenerately to different loaded patterns, resulting in many spurious states due to cross-talk effects. We present an idea to increase the number of attracting basins for patterns while suppressing the associated spurious states, by introducing coherent and collective firing in multispin groups. We numerically implement the method and test the number, stability, and basin size of the attractors thus created. Increasing the size of a group of coherent excitation suppresses spurious states, stabilizes loaded patterns, and dramatically increases the number of pattern attractors.

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50. Redundancy, Self-Motion, and Motor Control

V. Martin, J. P. Scholz, Gregor Schöner

https://doi.org/10.1162/neco.2008.01-08-698 | pp. 1371-1414

Outside the laboratory, human movement typically involves redundant effector systems. How the nervous system selects among the task-equivalent solutions may provide insights into how movement is controlled. We propose a process model of movement generation that accounts for the kinematics of goal-directed

pointing movements performed with a redundant arm. The key element is a neuronal dynamics that generates a virtual joint trajectory. This dynamics receives input from a neuronal timer that paces end-effector motion along its path. Within this dynamics, virtual joint velocity vectors that move the end effector are dynamically decoupled from velocity vectors that do not. Moreover, the sensed real joint configuration is coupled back into this neuronal dynamics, updating the virtual trajectory so that it yields to task-equivalent deviations from the dynamic movement plan. Experimental data from participants who perform in the same task setting as the model are compared in detail to the model predictions. We discover that joint velocities contain a substantial amount of self-motion that does not move the end effector. This is caused by the low impedance of muscle joint systems and by coupling among muscle joint systems due to multiarticulatory muscles. Back-coupling amplifies the induced control errors. We establish a link between the amount of self-motion and how curved the end-effector path is. We show that models in which an inverse dynamics cancels interaction torques predict too little self-motion and too straight end-effector paths.

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51. Analysis of Complete Stability for Discrete-Time Cellular Neural Networks with Piecewise Linear Output Functions

Xuemei Li

https://doi.org/10.1162/neco.2008.04-08-744 | pp. 1434-1458

This letter discusses the complete stability of discrete-time cellular neural networks with piecewise linear output functions. Under the assumption of certain symmetry on the feedback matrix, a sufficient condition of complete stability is derived by finite trajectory length. Because the symmetric conditions are not robust, the complete stability of networks may be lost under sufficiently small perturbations. The robust conditions of complete stability are also given for discrete-time cellular neural networks with multiple equilibrium points and a unique equilibrium point. These complete stability results are robust and available.

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52. Theory and Algorithm for Learning with Dissimilarity Functions

Liwei Wang, Masashi Sugiyama, Cheng Yang, Kohei Hatano, Jufu Feng

https://doi.org/10.1162/neco.2008.08-06-805 | pp. 1459-1484

We study the problem of classification when only a dissimilarity function between objects is accessible. That is, data samples are represented not by feature vectors but in terms of their pairwise dissimilarities. We establish sufficient conditions for dissimilarity functions to allow building accurate classifiers. The theory immediately suggests a learning paradigm: construct an ensemble of simple classifiers, each depending on a pair of examples; then find a convex combination of them to achieve a large margin. We next develop a practical algorithm referred to as dissimilarity-based boosting (DBoost) for learning with dissimilarity functions under theoretical guidance. Experiments on a variety of databases demonstrate that the DBoost algorithm is promising for several dissimilarity measures widely used in practice.

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53. Nonlinear Extraction of Independent Components of Natural Images Using Radial Gaussianization

Siwei Lyu, Eero P. Simoncelli

https://doi.org/10.1162/neco.2009.04-08-773 | pp. 1485-1519

We consider the problem of efficiently encoding a signal by transforming it to a new representation whose components are statistically independent. A widely studied linear solution, known as independent component analysis (ICA), exists for the case when the signal is generated as a linear transformation of independent nongaussian sources. Here, we examine a complementary case, in which the source is nongaussian and elliptically symmetric. In this case, no invertible linear transform suffices to decompose the signal into independent components, but we show that a simple nonlinear transformation, which we call radial gaussianization (RG), is able to remove all dependencies. We then examine this methodology in the context of natural image statistics. We first show that distributions of spatially proximal bandpass filter responses are better described as elliptical than as linearly transformed independent sources. Consistent with this, we demonstrate that the reduction in dependency achieved by applying RG to either nearby pairs or blocks of bandpass filter responses is significantly greater than that achieved by ICA. Finally, we show that the RG transformation may be closely approximated by divisive normalization, which has been used to model the nonlinear response properties of visual neurons.

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54. Dynamical Analysis of Bayesian Inference Models for the Eriksen Task

Yuan Sophie Liu, Angela J. Yu, Philip Holmes

https://doi.org/10.1162/neco.2009.03-07-495 | pp. 1520-1553

The Eriksen task is a classical paradigm that explores the effects of competing sensory inputs on response tendencies and the nature of selective attention in controlling these processes. In this task, conflicting flanker stimuli interfere with the processing of a central target, especially on short reaction time trials. This task has been modeled by neural networks and more recently by a normative Bayesian account. Here, we analyze the dynamics of the Bayesian models, which are nonlinear, coupled discrete time dynamical systems, by considering simplified, approximate systems that are linear and decoupled. Analytical solutions of these allow us to describe how posterior probabilities and psychometric functions depend on model parameters. We compare our results with numerical simulations of the original models and derive fits to experimental data, showing that agreements are rather good. We also investigate continuum limits of these simplified dynamical systems and demonstrate that Bayesian updating is closely related to a drift-diffusion process, whose implementation in neural network models has been extensively studied. This provides insight into how neural substrates can implement Bayesian computations.

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55. A First-Order Nonhomogeneous Markov Model for the Response of Spiking Neurons Stimulated by Small Phase-Continuous Signals

Jonathan Tapson, Craig Jin, André van Schaik, Ralph Etienne■Cummings

https://doi.org/10.1162/neco.2009.06-07-548 | pp. 1554-1588

We present a first-order nonhomogeneous Markov model for the interspike-interval density of a continuously stimulated spiking neuron. The model allows the conditional interspike-interval density and the stationary interspike-interval density to be expressed as products of two separate functions, one of which describes only the neuron characteristics and the other of which describes only the signal characteristics. The approximation shows particularly clearly that signal autocorrelations and cross-correlations arise as natural

features of the interspike-interval density and are particularly clear for small signals and moderate noise. We show that this model simplifies the design of spiking neuron cross-correlation systems and describe a four-neuron mutual inhibition network that generates a cross-correlation output for two input signals.

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56. A Binary Variable Model for Affinity Propagation

Inmar E. Givoni, Brendan J. Frey

https://doi.org/10.1162/neco.2009.05-08-785 | pp. 1589-1600

Affinity propagation (AP) was recently introduced as an unsupervised learning algorithm for exemplar-based clustering. We present a derivation of AP that is much simpler than the original one and is based on a quite different graphical model. The new model allows easy derivations of message updates for extensions and modifications of the standard AP algorithm. We demonstrate this by adjusting the new AP model to represent the capacitated clustering problem. For those wishing to investigate or extend the graphical model of the AP algorithm, we suggest using this new formulation since it allows a simpler and more intuitive model manipulation.

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57. A New Hypothesis for Sleep: Tuning for Criticality

Barak A. Pearlmutter, Conor Houghton

https://doi.org/10.1162/neco.2009.05-08-787 | pp. 1622-1641

We propose that the critical function of sleep is to prevent uncontrolled neuronal feedback while allowing rapid responses and prolonged retention of short-term memories. Through learning, the brain is tuned to react optimally to environmental challenges. Optimal behavior often requires rapid responses and the prolonged retention of short-term memories. At a neuronal level, these correspond to recurrent activity in local networks. Unfortunately, when a network exhibits recurrent activity, small changes in the parameters or conditions can lead to runaway oscillations. Thus, the very changes that improve the processing performance of the network can put it at risk of runaway oscillation. To prevent this, stimulus-dependent network changes should be permitted only when there is a margin of safety around the current network parameters. We propose that the essential role of sleep is to establish this margin by exposing the network to a variety of inputs, monitoring for erratic behavior, and adjusting the parameters. When sleep is not possible, an emergency mechanism must come into play, preventing runaway behavior at the expense of processing efficiency. This is tiredness.

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58. Generation of Spike Trains with Controlled Auto- and Cross-Correlation Functions

Michael Krumin, Shy Shoham

https://doi.org/10.1162/neco.2009.08-08-847 | pp. 1642-1664

Emerging evidence indicates that information processing, as well as learning and memory processes, in both the network and single-neuron levels are highly dependent on the correlation structure of multiple spike trains. Contemporary experimental as well as theoretical studies that involve quasi-realistic neuronal stimulation thus require a method for controlling spike train correlations. This letter introduces a general new

strategy for generating multiple spike trains with exactly controlled mean firing rates and correlation structure (defined in terms of auto- and cross-correlation functions). Our approach nonlinearly transforms random gaussian-distributed processes with a predistorted correlation structure into nonnegative rate processes, which are then used to generate doubly stochastic Poisson point processes with the required correlation structure. We show how this approach can be used to generate stationary or nonstationary spike trains from small or large groups of neurons with diverse auto- and cross-correlation structures. We analyze and derive analytical formulas for the high-order correlation structure of generated spike trains and discuss the limitations of this approach.

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59. How Does Maintenance of Network Activity Depend on Endogenous Dynamics of Isolated Neurons?

Andrey V. Olypher, Ronald L. Calabrese

https://doi.org/10.1162/neco.2009.01-08-685 | pp. 1665-1682

Robust activity of some networks, such as central pattern generators, suggests the existence of physiological mechanisms that maintain the most important characteristics, for example, the period and spike frequency of the pattern. Whatever these mechanisms are, they change the appropriate model parameters to or along the isomanifolds on which the characteristics of the pattern are constant, while their sensitivities to parameters may be different. Setting synaptic connections to zero at the points of isomanifolds allows for dissecting the maintenance mechanisms into components involving synaptic transmission and components involving intrinsic currents. The physiological meaning of the intrinsic current changes might be revealed by analysis of their impact on endogenous neuronal dynamics. Here, we sought answers to two questions: (1) Do parameter variations in insensitive directions (along isomanifolds) change endogenous dynamics of the network neurons? (2) Do sensitive and insensitive directions for network pattern characteristics depend on endogenous dynamics of the network neurons? We considered a leech heartbeat half-center oscillator model network and analyzed isomanifolds on which the burst period or spike frequency of the model, or both, are constant. Based on our analysis, we hypothesize that the dependence on endogenous dynamics of the isolated neurons is the stronger the more characteristics of the pattern have to be maintained. We also found that in general, the network was more flexible when it consisted of endogenously tonically spiking rather than bursting or silent neurons. Finally, we discuss the physiological implications of our findings.

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60. GABA Transporter Preserving Ongoing Spontaneous Neuronal Activity at Firing Subthreshold

Osamu Hoshino

https://doi.org/10.1162/neco.2009.05-08-778 | pp. 1683-1713

There has been compelling evidence that the GABA transporter is crucial not only for removing gamma-aminobutyric acid (GABA) from but also releasing it into extracellular space, thereby clamping ambient GABA (GABA in extracellular space) at a certain level. The ambient GABA is known to activate extrasynaptic GABA receptors and provide tonic inhibitory current into neurons. We investigated how the transporter regulates the level of ambient GABA, mediates tonic neuronal inhibition, and influences ongoing spontaneous neuronal activity. A cortical neural network model is proposed in which GABA transporters on

lateral (L) and feedback (F) inhibitory (GABAergic) interneurons are functionally made. Principal (P) cell assemblies participate in expressing information about elemental sensory features. At membrane potentials below the reversal potential, there is net influx of GABA, whereas at membrane potentials above the reversal potential, there is net efflux of GABA. Through this transport mechanism, ambient GABA concentration is kept within a submicromolar range during an ongoing spontaneous neuronal activity time period. Here we show that the GABA transporter on L cells regulates the overall level of ambient GABA across cell assemblies, and that on F cells it does so within individual cell assemblies. This combinatorial regulation of ambient GABA allows P cells to oscillate near firing threshold during the ongoing time period, thereby reducing their reaction time to externally applied stimuli. We suggest that the GABA transporter, with its forward and reverse transport mechanism, could regulate the ambient GABA. This transporter-mediated ambient GABA regulation may contribute to establishing an ongoing subthreshold neuronal state by which the network can respond rapidly to subsequent sensory input.

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61. Capacity of a Single Spiking Neuron Channel

Shiro Ikeda, Jonathan H. Manton

https://doi.org/10.1162/neco.2009.05-08-792 | pp. 1714-1748

Information transfer through a single neuron is a fundamental component of information processing in the brain, and computing the information channel capacity is important to understand this information processing. The problem is difficult since the capacity depends on coding, characteristics of the communication channel, and optimization over input distributions, among other issues. In this letter, we consider two models. The temporal coding model of a neuron as a communication channel assumes the output is tau where tau is a gamma-distributed random variable corresponding to the interspike interval, that is, the time it takes for the neuron to fire once. The rate coding model is similar; the output is the actual rate of firing over a fixed period of time. Theoretical studies prove that the distribution of inputs, which achieves channel capacity, is a discrete distribution with finite mass points for temporal and rate coding under a reasonable assumption. This allows us to compute numerically the capacity of a neuron. Numerical results are in a plausible range based on biological evidence to date.

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62. Adaptive Synchronization of Activities in a Recurrent Network

Thomas Voegtlin

https://doi.org/10.1162/neco.2009.02-08-708 | pp. 1749-1775

Predictive learning rules, where synaptic changes are driven by the difference between a random input and its reconstruction derived from internal variables, have proven to be very stable and efficient. However, it is not clear how such learning rules could take place in biological synapses. Here we propose an implementation that exploits the synchronization of neural activities within a recurrent network. In this framework, the asymmetric shape of spike-timing-dependent plasticity (STDP) can be interpreted as a self-stabilizing mechanism. Our results suggest a novel hypothesis concerning the computational role of neural synchrony and oscillations.

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63. Discrete- and Continuous-Time Probabilistic Models and Algorithms for Inferring Neuronal UP and DOWN States

Zhe Chen, Sujith Vijayan, Riccardo Barbieri, Matthew Wilson, Emery N. Brown

https://doi.org/10.1162/neco.2009.06-08-799 | pp. 1797-1862

UP and DOWN states, the periodic fluctuations between increased and decreased spiking activity of a neuronal population, are a fundamental feature of cortical circuits. Understanding UP-DOWN state dynamics is important for understanding how these circuits represent and transmit information in the brain. To date, limited work has been done on characterizing the stochastic properties of UP-DOWN state dynamics. We present a set of Markov and semi-Markov discrete- and continuous-time probability models for estimating UP and DOWN states from multiunit neural spiking activity. We model multiunit neural spiking activity as a stochastic point process, modulated by the hidden (UP and DOWN) states and the ensemble spiking history. We estimate jointly the hidden states and the model parameters by maximum likelihood using an expectation-maximization (EM) algorithm and a Monte Carlo EM algorithm that uses reversible-jump Markov chain Monte Carlo sampling in the E-step. We apply our models and algorithms in the analysis of both simulated multiunit spiking activity and actual multi- unit spiking activity recorded from primary somatosensory cortex in a behaving rat during slow-wave sleep. Our approach provides a statistical characterization of UP-DOWN state dynamics that can serve as a basis for verifying and refining mechanistic descriptions of this process.

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64. Maximally Reliable Markov Chains Under Energy Constraints

G. Sean Escola, Michael Eisele, Kenneth D. Miller, Liam Paninski

https://doi.org/10.1162/neco.2009.08-08-843 | pp. 1863-1912

Signal-to-noise ratios in physical systems can be significantly degraded if the outputs of the systems are highly variable. Biological processes for which highly stereotyped signal generations are necessary features appear to have reduced their signal variabilities by employing multiple processing steps. To better understand why this multistep cascade structure might be desirable, we prove that the reliability of a signal generated by a multistate system with no memory (i.e., a Markov chain) is maximal if and only if the system topology is such that the process steps irreversibly through each state, with transition rates chosen such that an equal fraction of the total signal is generated in each state. Furthermore, our result indicates that by increasing the number of states, it is possible to arbitrarily increase the reliability of the system. In a physical system, however, an energy cost is associated with maintaining irreversible transitions, and this cost increases with the number of such transitions (i.e., the number of states). Thus, an infinite-length chain, which would be perfectly reliable, is infeasible. To model the effects of energy demands on the maximally reliable solution, we numerically optimize the topology under two distinct energy functions that penalize either irreversible transitions or incommunicability between states, respectively. In both cases, the solutions are essentially irreversible linear chains, but with upper bounds on the number of states set by the amount of available energy. We therefore conclude that a physical system for which signal reliability is important should employ a linear architecture, with the number of states (and thus the reliability) determined by the intrinsic energy constraints of the system.

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65. Generalization of the Event-Based Carnevale-Hines Integration Scheme for Integrate-and-Fire Models

Ronald A. J. van Elburg, Arjen van Ooyen

https://doi.org/10.1162/neco.2009.07-08-815 | pp. 1913-1930

An event-based integration scheme for an integrate-and-fire neuron model with exponentially decaying excitatory synaptic currents and double exponential inhibitory synaptic currents has been introduced by Carnevale and Hines. However, the integration scheme imposes nonphysiological constraints on the time constants of the synaptic currents, which hamper its general applicability. This letter addresses this problem in two ways. First, we provide physical arguments demonstrating why these constraints on the time constants can be relaxed. Second, we give a formal proof showing which constraints can be abolished. As part of our formal proof, we introduce the generalized Carnevale-Hines lemma, a new tool for comparing double exponentials as they naturally occur in many cascaded decay systems, including receptor-neurotransmitter dissociation followed by channel closing. Through repeated application of the generalized lemma, we lift most of the original constraints on the time constants. Thus, we show that the Carnevale-Hines integration scheme for the integrate-and-fire model can be employed for simulating a much wider range of neuron and synapse types than was previously thought.

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66. Estimating Instantaneous Irregularity of Neuronal Firing

Takeaki Shimokawa, Shigeru Shinomoto

https://doi.org/10.1162/neco.2009.08-08-841 | pp. 1931-1951

Cortical neurons in vivo had been regarded as Poisson spike generators that convey no information other than the rate of random firing. Recently, using a metric for analyzing local variation of interspike intervals, researchers have found that individual neurons express specific patterns in generating spikes, which may symbolically be termed regular, random, or bursty, rather invariantly in time. In order to study the dynamics of firing patterns in greater detail, we propose here a Bayesian method for estimating firing irregularity and the firing rate simultaneously for a given spike sequence, and we implement an algorithm that may render the empirical Bayesian estimation practicable for data comprising a large number of spikes. Application of this method to electrophysiological data revealed a subtle correlation between the degree of firing irregularity and the firing rate for individual neurons. Irregularity of firing did not deviate greatly around the low degree of dependence on the firing rate and remained practically unchanged for individual neurons in the cortical areas V1 and MT, whereas it fluctuated greatly in the lateral geniculate nucleus of the thalamus. This indicates the presence and absence of autocontrolling mechanisms for maintaining patterns of firing in the cortex and thalamus, respectively.

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67. Combining Feature- and Correspondence-Based Methods for Visual Object Recognition

Günter Westphal, Rolf P. Würtz

https://doi.org/10.1162/neco.2009.12-07-675 | pp. 1952-1989

We present an object recognition system built on a combination of feature- and correspondence-based pattern recognizers. The feature-based part, called preselection network, is a single-layer feedforward

network weighted with the amount of information contributed by each feature to the decision at hand. For processing arbitrary objects, we employ small, regular graphs whose nodes are attributed with Gabor amplitudes, termed parquet graphs. The preselection network can quickly rule out most irrelevant matches and leaves only the ambiguous cases, so-called model candidates, to be verified by a rudimentary version of elastic graph matching, a standard correspondence-based technique for face and object recognition. According to the model, graphs are constructed that describe the object in the input image well. We report the results of experiments on standard databases for object recognition. The method achieved high recognition rates on identity and pose. Unlike many other models, it can also cope with varying background, multiple objects, and partial occlusion.

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68. Stochastic Variance Models in Discrete Time with Feedforward Neural Networks

Charles Andoh

https://doi.org/10.1162/neco.2009.11-07-642 | pp. 1990-2008

The study overcomes the estimation difficulty in stochastic variance models for discrete financial time series with feedforward neural networks. The volatility function is estimated semiparametrically. The model is used to estimate market risk, taking into account not only the time series of interest but extra information on the market. As an application, some stock prices series are studied and compared with the nonlinear ARX-ARCHX model.

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69. An Internal Model for Acquisition and Retention of Motor Learning During Arm Reaching

Luca Lonini, Laura Dipietro, Loredana Zollo, Eugenio Guglielmelli, Hermano Igo Krebs

https://doi.org/10.1162/neco.2009.03-08-721 | pp. 2009-2027

Humans have the ability to learn novel motor tasks while manipulating the environment. Several models of motor learning have been proposed in the literature, but few of them address the problem of retention and interference of motor memory. The modular selection and identification for control (MOSAIC) model, originally proposed by Wolpert and Kawato, is one of the most relevant contributions; it suggests a possible strategy on how the human motor control system learns and adapts to novel environments. MOSAIC employs the concept of forward and inverse models. The same group later proposed the hidden Markov model (HMM) MOSAIC, which affords learning multiple tasks. The significant drawback of this second approach is that the HMM must be trained with a complete data set that includes all contexts. Since the number of contexts or modules is fixed from the onset, this approach does not afford incremental learning of new tasks. In this letter, we present an alternative architecture to overcome this problem, based on a nonparametric regression algorithm, named locally weighted projection regression (LWPR). This network structure develops according to the contexts allowing incremental training. Of notice, interaction force is used to disambiguate among different contexts. We demonstrate the capability of this alternative architecture with a simulated 2 degree-of-freedom representation of the human arm that learns to interact with three distinct objects, reproducing the same test paradigm of the HMM MOSAIC. After learning the dynamics of the three objects, the LWPR network successfully learns to compensate for a novel velocity-dependent force field. Equally important, it retains previously acquired knowledge on the interaction

with the three objects. Thus, this architecture allows both incremental learning of new tasks and retention of previously acquired knowledge, a feature of human motor learning and memory.

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70. Estimation of Positive Semidefinite Correlation Matrices by Using Convex Quadratic Semidefinite Programming

Tadayoshi Fushiki

https://doi.org/10.1162/neco.2009.04-08-765 | pp. 2028-2048

The correlation matrix is a fundamental statistic that is used in many fields. For example, GroupLens, a collaborative filtering system, uses the correlation between users for predictive purposes. Since the correlation is a natural similarity measure between users, the correlation matrix may be used in the Gram matrix in kernel methods. However, the estimated correlation matrix sometimes has a serious defect: although the correlation matrix is originally positive semidefinite, the estimated one may not be positive semidefinite when not all ratings are observed. To obtain a positive semidefinite correlation matrix, the nearest correlation matrix problem has recently been studied in the fields of numerical analysis and optimization. However, statistical properties are not explicitly used in such studies. To obtain a positive semidefinite correlation matrix, we assume the approximate model. By using the model, an estimate is obtained as the optimal point of an optimization problem formulated with information on the variances of the estimated correlation coefficients. The problem is solved by a convex quadratic semidefinite program. A penalized likelihood approach is also examined. The MovieLens data set is used to test our approach.

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71. A Multiclass Classification Method Based on Decoding of Binary Classifiers

Takashi Takenouchi, Shin Ishii

https://doi.org/10.1162/neco.2009.03-08-740 | pp. 2049-2081

In this letter, we present new methods of multiclass classification that combine multiple binary classifiers. Misclassification of each binary classifier is formulated as a bit inversion error with probabilistic models by making an analogy to the context of information transmission theory. Dependence between binary classifiers is incorporated into our model, which makes a decoder a type of Boltzmann machine. We performed experimental studies using a synthetic data set, data sets from the UCI repository, and bioinformatics data sets, and the results show that the proposed methods are superior to the existing multiclass classification methods.

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72. Validation-Based Sparse Gaussian Process Classifier Design

Shirish Shevade, S. Sundararajan

https://doi.org/10.1162/neco.2009.03-08-724 | pp. 2082-2103

Gaussian processes (GPs) are promising Bayesian methods for classification and regression problems. Design of a GP classifier and making predictions using it is, however, computationally demanding, especially when the training set size is large. Sparse GP classifiers are known to overcome this limitation. In this letter, we propose and study a validation-based method for sparse GP classifier design. The proposed

method uses a negative log predictive (NLP) loss measure, which is easy to compute for GP models. We use this measure for both basis vector selection and hyperparameter adaptation. The experimental results on several real-world benchmark data sets show better or comparable generalization performance over existing methods.

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73. Direct Estimation of Inhomogeneous Markov Interval Models of Spike Trains

Daniel K. Wójcik, Gabriela Mochol, Wit Jakuczun, Marek Wypych, Wioletta J. Waleszczyk

https://doi.org/10.1162/neco.2009.07-08-828 | pp. 2105-2113

A necessary ingredient for a quantitative theory of neural coding is appropriate "spike kinematics": a precise description of spike trains. While summarizing experiments by complete spike time collections is clearly inefficient and probably unnecessary, the most common probabilistic model used in neurophysiology, the inhomogeneous Poisson process, often seems too crude. Recently a more general model, the inhomogeneous Markov interval model (Berry & Meister, 1998 ; Kass & Ventura, 2001), was considered, which takes into account both the current experimental time and the time from the last spike. Several techniques were proposed to estimate the parameters of these models from data. Here we propose a direct method of estimation that is easy to implement, fast, and conceptually simple. The method is illustrated with an analysis of sample data from the cat's superior colliculus.

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74. Importance of the Cutoff Value in the Quadratic Adaptive Integrate-and-Fire Model

Jonathan Touboul

https://doi.org/10.1162/neco.2009.09-08-853 | pp. 2114-2122

The quadratic adaptive integrate-and-fire model (Izhikevich, 2003 , 2007) is able to reproduce various firing patterns of cortical neurons and is widely used in large-scale simulations of neural networks. This model describes the dynamics of the membrane potential by a differential equation that is quadratic in the voltage, coupled to a second equation for adaptation. Integration is stopped during the rise phase of a spike at a voltage cutoff value V(c) or when it blows up. Subsequently the membrane potential is reset, and the adaptation variable is increased by a fixed amount. We show in this note that in the absence of a cutoff value, not only the voltage but also the adaptation variable diverges in finite time during spike generation in the quadratic model. The divergence of the adaptation variable makes the system very sensitive to the cutoff: changing V(c) can dramatically alter the spike patterns. Furthermore, from a computational viewpoint, the divergence of the adaptation variable implies that the time steps for numerical simulation need to be small and adaptive. However, divergence of the adaptation variable does not occur for the quartic model (Touboul, 2008) and the adaptive exponential integrate-and-fire model (Brette & Gerstner, 2005). Hence, these models are robust to changes in the cutoff value.

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75. Fast and Robust Learning by Reinforcement Signals: Explorations in the Insect Brain

Ramón Huerta, Thomas Nowotny

https://doi.org/10.1162/neco.2009.03-08-733 | pp. 2123-2151

We propose a model for pattern recognition in the insect brain. Departing from a well-known body of knowledge about the insect brain, we investigate which of the potentially present features may be useful to learn input patterns rapidly and in a stable manner. The plasticity underlying pattern recognition is situated in the insect mushroom bodies and requires an error signal to associate the stimulus with a proper response. As a proof of concept, we used our model insect brain to classify the well-known MNIST database of handwritten digits, a popular benchmark for classifiers. We show that the structural organization of the insect brain appears to be suitable for both fast learning of new stimuli and reasonable performance in stationary conditions. Furthermore, it is extremely robust to damage to the brain structures involved in sensory processing. Finally, we suggest that spatiotemporal dynamics can improve the level of confidence in a classification decision. The proposed approach allows testing the effect of hypothesized mechanisms rather than speculating on their benefit for system performance or confidence in its responses.

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76. Quantifying Statistical Interdependence by Message Passing on Graphs—Part I: One-Dimensional Point Processes

Justin Dauwels, François Vialatte, Théophane Weber, Andrzej Cichocki

https://doi.org/10.1162/neco.2009.04-08-746 | pp. 2152-2202

We present a novel approach to quantify the statistical interdependence of two time series, referred to as stochastic event synchrony (SES). The first step is to extract the two given time series. The next step is to try to align events from one time series with events from the other. The better the alignment the more similar the two series are considered to be. More precisely, the similarity is quantified by the following parameters: time delay, variance of the time jitter, fraction of noncoincident events, and average similarity of the aligned events. The pairwise alignment and SES parameters are determined by statistical inference. In particular, the SES parameters are computed by maximum a posteriori (MAP) estimation, and the pairwise alignment is obtained by applying the max product algorithm. This letter deals with one-dimensional point processes; the extension to multidimensional point processes is considered in a companion letter in this issue. By analyzing surrogate data, we demonstrate that SES is able to quantify both timing precision and event reliability more robustly than classical measures can. As an illustration, neuronal spike data generated by Morris-Lecar neuron model are considered.

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77. Quantifying Statistical Interdependence by Message Passing on Graphs—Part II: Multidimensional Point Processes

Justin Dauwels, François Vialatte, Théophane Weber, Toshimitsu Musha, Andrzej Cichocki

https://doi.org/10.1162/neco.2009.11-08-899 | pp. 2203-2268

Stochastic event synchrony is a technique to quantify the similarity of pairs of signals. First, events are extracted from the two given time series. Next, one tries to align events from one time series with events from the other. The better the alignment, the more similar the two time series are considered to be. In Part I, the companion letter in this issue, one-dimensional events are considered; this letter concerns multidimensional events. Although the basic idea is similar, the extension to multidimensional point

processes involves a significantly more difficult combinatorial problem and therefore is nontrivial. Also in the multidimensional case, the problem of jointly computing the pairwise alignment and SES parameters is cast as a statistical inference problem. This problem is solved by coordinate descent, more specifically, by alternating the following two steps: (1) estimate the SES parameters from a given pairwise alignment; (2) with the resulting estimates, refine the pairwise alignment. The SES parameters are computed by maximum a posteriori (MAP) estimation (step 1), in analogy to the one-dimensional case. The pairwise alignment (step 2) can no longer be obtained through dynamic programming, since the state space becomes too large. Instead it is determined by applying the max-product algorithm on a cyclic graphical model. In order to test the robustness and reliability of the SES method, it is first applied to surrogate data. Next, it is applied to detect anomalies in EEG synchrony of mild cognitive impairment (MCI) patients. Numerical results suggest that SES is significantly more sensitive to perturbations in EEG synchrony than a large variety of classical synchrony measures.

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78. Stimulus-Dependent Correlations in Threshold-Crossing Spiking Neurons

Yoram Burak, Sam Lewallen, Haim Sompolinsky

https://doi.org/10.1162/neco.2009.07-08-830 | pp. 2269-2308

We consider a threshold-crossing spiking process as a simple model for the activity within a population of neurons. Assuming that these neurons are driven by a common fluctuating input with gaussian statistics, we evaluate the cross-correlation of spike trains in pairs of model neurons with different thresholds. This correlation function tends to be asymmetric in time, indicating a preference for the neuron with the lower threshold to fire before the one with the higher threshold, even if their inputs are identical. The relationship between these results and spike statistics in other models of neural activity is explored. In particular, we compare our model with an integrate-and-fire model in which the membrane voltage resets following each spike. The qualitative properties of spike cross-correlations, emerging from the threshold-crossing model, are similar to those of bursting events in the integrate-and-fire model. This is particularly true for generalized integrate-and-fire models in which spikes tend to occur in bursts, as observed, for example, in retinal ganglion cells driven by a rapidly fluctuating visual stimulus. The threshold-crossing model thus provides a simple, analytically tractable description of event onsets in these neurons.

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79. Information-Geometric Measures as Robust Estimators of Connection Strengths and External Inputs

Masami Tatsuno, Jean■Marc Fellous, ■■■-■■■ ■■■■■

https://doi.org/10.1162/neco.2009.04-08-748 | pp. 2309-2335

Information geometry has been suggested to provide a powerful tool for analyzing multineuronal spike trains. Among several advantages of this approach, a significant property is the close link between information-geometric measures and neural network architectures. Previous modeling studies established that the first- and second-order information-geometric measures corresponded to the number of external inputs and the connection strengths of the network, respectively. This relationship was, however, limited to a symmetrically connected network, and the number of neurons used in the parameter estimation of the log-linear model needed to be known. Recently, simulation studies of biophysical model neurons have suggested that information geometry can estimate the relative change of connection strengths and external

inputs even with asymmetric connections. Inspired by these studies, we analytically investigated the link between the information-geometric measures and the neural network structure with asymmetrically connected networks of N neurons. We focused on the information-geometric measures of orders one and two, which can be derived from the two-neuron log-linear model, because unlike higher-order measures, they can be easily estimated experimentally. Considering the equilibrium state of a network of binary model neurons that obey stochastic dynamics, we analytically showed that the corrected first- and second-order information-geometric measures provided robust and consistent approximation of the external inputs and connection strengths, respectively. These results suggest that information-geometric measures provide useful insights into the neural network architecture and that they will contribute to the study of system-level neuroscience.

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80. Time-Varying Perturbations Can Distinguish Among Integrate-to-Threshold Models for Perceptual Decision Making in Reaction Time Tasks

Xiang Zhou, KongFatt Wong■Lin, Holmes Philip

https://doi.org/10.1162/neco.2009.07-08-817 | pp. 2336-2362

Several integrate-to-threshold models with differing temporal integration mechanisms have been proposed to describe the accumulation of sensory evidence to a prescribed level prior to motor response in perceptual decision-making tasks. An experiment and simulation studies have shown that the introduction of time-varying perturbations during integration may distinguish among some of these models. Here, we present computer simulations and mathematical proofs that provide more rigorous comparisons among one-dimensional stochastic differential equation models. Using two perturbation protocols and focusing on the resulting changes in the means and standard deviations of decision times, we show that for high signal-to-noise ratios, drift-diffusion models with constant and time-varying drift rates can be distinguished from Ornstein-Uhlenbeck processes, but not necessarily from each other. The protocols can also distinguish stable from unstable Ornstein-Uhlenbeck processes, and we show that a nonlinear integrator can be distinguished from these linear models by changes in standard deviations. The protocols can be implemented in behavioral experiments.

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81. Randomized Shortest-Path Problems: Two Related Models

Marco Saerens, Youssef Achbany, François Fouss, Luh Yen

https://doi.org/10.1162/neco.2009.11-07-643 | pp. 2363-2404

This letter addresses the problem of designing the transition probabilities of a finite Markov chain (the policy) in order to minimize the expected cost for reaching a destination node from a source node while maintaining a fixed level of entropy spread throughout the network (the exploration). It is motivated by the following scenario. Suppose you have to route agents through a network in some optimal way, for instance, by minimizing the total travel cost-nothing particular up to now-you could use a standard shortest-path algorithm. Suppose, however, that you want to avoid pure deterministic routing policies in order, for instance, to allow some continual exploration of the network, avoid congestion, or avoid complete predictability of your routing strategy. In other words, you want to introduce some randomness or unpredictability in the routing policy (i.e., the routing policy is randomized). This problem, which will be called

the randomized shortest-path problem (RSP), is investigated in this work. The global level of randomness of the routing policy is quantified by the expected Shannon entropy spread throughout the network and is provided a priori by the designer. Then, necessary conditions to compute the optimal randomized policy-minimizing the expected routing cost-are derived. Iterating these necessary conditions, reminiscent of Bellman's value iteration equations, allows computing an optimal policy, that is, a set of transition probabilities in each node. Interestingly and surprisingly enough, this first model, while formulated in a totally different framework, is equivalent to Akamatsu's model (1996), appearing in transportation science, for a special choice of the entropy constraint. We therefore revisit Akamatsu's model by recasting it into a sum-over-paths statistical physics formalism allowing easy derivation of all the quantities of interest in an elegant, unified way. For instance, it is shown that the unique optimal policy can be obtained by solving a simple linear system of equations. This second model is therefore more convincing because of its computational efficiency and soundness. Finally, simulation results obtained on simple, illustrative examples show that the models behave as expected.

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82. Erratum (“How Does Maintenance of Network Activity Depend on Endogenous Dynamics of Isolated Neurons?” by Andrey V. Olypher and Ronald L. Calabrese, Neural Computation, June 2009, Vol. 21, No. 6: 1665–1682)

(authors unknown)

https://doi.org/10.1162/neco.2009.21.8.2405b | pp. 2405-2405

In the June 2009 issue of Neural Computation (Volume 21, Number 6), the letter “How Does Maintenance of Network Activity Depend on Endogenous Dynamics of Isolated Neurons?” by Andrey V. Olypher and Ronald L. Calabrese was printed with the incorrect communicator name listed.Printed version reads:Toby BergerThe correct communicator name should read:Kevin Briggmann

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83. Erratum (“A First-Order Nonhomogeneous Markov Model for the Response of Spiking Neurons Stimulated by Small Phase-Continuous Signals” by Jonathan Tapson, Craig Jin, André van Schaik, and Ralph Etienne-Cummings, Neural Computation, June 2009, Vol. 21, No. 6: 1554–1588)

(authors unknown)

https://doi.org/10.1162/neco.2009.21.8.2405a | pp. 2405-2405

In the June 2009 issue of Neural Computation (Volume 21, Number 6), the article “A First-Order Nonhomogeneous Markov Model for the Response of Spiking Neurons Stimulated by Small Phase-Continuous Signals” by Jonathan Tapson, Craig Jin, André van Schaik, and Ralph Etienne-Cummings was printed with the incorrect accepted date.Printed version reads:Accepted December 29, 2007The correct date should read November 18, 2008.

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84. Sequential Effects in Two-Choice Reaction Time Tasks: Decomposition and Synthesis of Mechanisms

Juan Gao, KongFatt Wong■Lin, Philip Holmes, Patrick Simen, Jonathan D. Cohen

https://doi.org/10.1162/neco.2009.09-08-866 | pp. 2407-2436

Performance on serial tasks is influenced by first- and higher-order sequential effects, respectively, due to the immediately previous and earlier trials. As response-to-stimulus interval (RSI) increases, the pattern of reaction times transits from a benefit-only mode, traditionally ascribed to automatic facilitation (AF), to a cost-benefit mode, due to strategic expectancy (SE). To illuminate the sources of such effects, we develop a connectionist network of two mutually inhibiting neural decision units subject to feedback from previous trials. A study of separate biasing mechanisms shows that residual decision unit activity can lead to only first-order AF, but higher-order AF can result from strategic priming mediated by conflict monitoring, which we instantiate in two distinct versions. A further mechanism mediates expectation-related biases that grow during RSI toward saturation levels determined by weighted repetition (or alternation) sequence lengths. Equipped with these mechanisms, the network, consistent with known neurophysiology, accounts for several sets of behavioral data over a wide range of RSIs. The results also suggest that practice speeds up all the mechanisms rather than adjusting their relative strengths.

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85. Computation with Spikes in a Winner-Take-All Network

Matthias Oster, Rodney J. Douglas, Shih■Chii Liu

https://doi.org/10.1162/neco.2009.07-08-829 | pp. 2437-2465

The winner-take-all (WTA) computation in networks of recurrently connected neurons is an important decision element of many models of cortical processing. However, analytical studies of the WTA performance in recurrent networks have generally addressed rate-based models. Very few have addressed networks of spiking neurons, which are relevant for understanding the biological networks themselves and also for the development of neuromorphic electronic neurons that commmunicate by action potential like address-events. Here, we make steps in that direction by using a simplified Markov model of the spiking network to examine analytically the ability of a spike-based WTA network to discriminate the statistics of inputs ranging from stationary regular to nonstationary Poisson events. Our work extends previous theoretical results showing that a WTA recurrent network receiving regular spike inputs can select the correct winner within one interspike interval. We show first for the case of spike rate inputs that input discrimination and the effects of self-excitation and inhibition on this discrimination are consistent with results obtained from the standard rate-based WTA models. We also extend this discrimination analysis of spiking WTAs to nonstationary inputs with time-varying spike rates resembling statistics of real-world sensory stimuli. We conclude that spiking WTAs are consistent with their continuous counterparts for steady-state inputs, but they also exhibit high discrimination performance with nonstationary inputs.

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86. Automatic Spike Sorting Using Tuning Information

Valérie Ventura

https://doi.org/10.1162/neco.2009.12-07-669 | pp. 2466-2501

Current spike sorting methods focus on clustering neurons' characteristic spike waveforms. The resulting spike-sorted data are typically used to estimate how covariates of interest modulate the firing rates of neurons. However, when these covariates do modulate the firing rates, they provide information about spikes' identities, which thus far have been ignored for the purpose of spike sorting. This letter describes a novel approach to spike sorting, which incorporates both waveform information and tuning information obtained from the modulation of firing rates. Because it efficiently uses all the available information, this spike sorter yields lower spike misclassification rates than traditional automatic spike sorters. This theoretical result is verified empirically on several examples. The proposed method does not require additional assumptions; only its implementation is different. It essentially consists of performing spike sorting and tuning estimation simultaneously rather than sequentially, as is currently done. We used an expectation-maximization maximum likelihood algorithm to implement the new spike sorter. We present the general form of this algorithm and provide a detailed implementable version under the assumptions that neurons are independent and spike according to Poisson processes. Finally, we uncover a systematic flaw of spike sorting based on waveform information only.

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87. Belief Propagation in Networks of Spiking Neurons

Andreas Steimer, Wolfgang Maass, Rodney J. Douglas

https://doi.org/10.1162/neco.2009.08-08-837 | pp. 2502-2523

From a theoretical point of view, statistical inference is an attractive model of brain operation. However, it is unclear how to implement these inferential processes in neuronal networks. We offer a solution to this problem by showing in detailed simulations how the belief propagation algorithm on a factor graph can be embedded in a network of spiking neurons. We use pools of spiking neurons as the function nodes of the factor graph. Each pool gathers "messages" in the form of population activities from its input nodes and combines them through its network dynamics. Each of the various output messages to be transmitted over the edges of the graph is computed by a group of readout neurons that feed in their respective destination pools. We use this approach to implement two examples of factor graphs. The first example, drawn from coding theory, models the transmission of signals through an unreliable channel and demonstrates the principles and generality of our network approach. The second, more applied example is of a psychophysical mechanism in which visual cues are used to resolve hypotheses about the interpretation of an object's shape and illumination. These two examples, and also a statistical analysis, demonstrate good agreement between the performance of our networks and the direct numerical evaluation of belief propagation.

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88. Stochastic Properties of Coincidence-Detector Neural Cells

Ram Krips, Miriam Furst

https://doi.org/10.1162/neco.2009.07-07-563 | pp. 2524-2553

Neural information is characterized by sets of spiking events that travel within the brain through neuron junctions that receive, transmit, and process streams of spikes. Coincidence detection is one of the ways to describe the functionality of a single neural cell. This letter presents an analytical derivation of the output stochastic behavior of a coincidence detector (CD) cell whose stochastic inputs behave as a nonhomogeneous Poisson process (NHPP) with both excitatory and inhibitory inputs. The derivation, which

is based on an efficient breakdown of the cell into basic functional elements, results in an output process whose behavior can be approximated as an NHPP as long as the coincidence interval is much smaller than the refractory period of the cell's inputs. Intuitively, the approximation is valid as long as the processing rate is much faster than the incoming information rate. This type of modeling is a simplified but very useful description of neurons since it enables analytical derivations. The statistical properties of single CD cell's output make it possible to integrate and analyze complex neural cells in a feedforward network using the methodology presented here. Accordingly, basic biological characteristics of neural activity are demonstrated, such as a decrease in the spontaneous rate at higher brain levels and improved signal-to-noise ratio for harmonic input signals.

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89. Constraint on the Number of Synaptic Inputs to a Visual Cortical Neuron Controls Receptive Field Formation

Shigeru Tanaka, Masanobu Miyashita

https://doi.org/10.1162/neco.2009.04-08-752 | pp. 2554-2580

To date, Hebbian learning combined with some form of constraint on synaptic inputs has been demonstrated to describe well the development of neural networks. The previous models revealed mathematically the importance of synaptic constraints to reproduce orientation selectivity in the visual cortical neurons, but biological mechanisms underlying such constraints remain unclear. In this study, we addressed this issue by formulating a synaptic constraint based on activity-dependent mechanisms of synaptic changes. Particularly, considering metabotropic glutamate receptor-mediated long-term depression, we derived synaptic constraint that suppresses the number of inputs from individual presynaptic neurons. We performed computer simulations of the activity-dependent self-organization of geniculocortical inputs with the synaptic constraint and examined the formation of receptive fields (RFs) of model visual cortical neurons. When we changed the magnitude of the synaptic constraint, we found the emergence of distinct RF structures such as concentric RFs, simple-cell-like RFs, and double-oriented RFs and also a gradual transition between spatiotemporal separable and inseparable RFs. Thus, the model based on the synaptic constraint derived from biological consideration can account systematically for the repertoire of RF structures observed in the primary visual cortices of different species for the first time.

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90. Orientation Disparity: A Cue for 3D Orientation?

Hal S. Greenwald, David C. Knill

https://doi.org/10.1162/neco.2009.08-08-848 | pp. 2581-2604

Orientation disparity, the difference in orientation that results when a texture element on a slanted surface is projected to the two eyes, has been proposed as a binocular cue for 3D orientation. Since orientation disparity is confounded with position disparity, neither behavioral nor neurophysiological experiments have successfully isolated its contribution to slant estimates or established whether the visual system uses it. Using a modified disparity energy model, we simulated a population of binocular visual cortical neurons tuned to orientation disparity and measured the amount of Fisher information contained in the activity patterns. We evaluated the potential contribution of orientation disparity to 3D orientation estimation and delimited the stimulus conditions under which it is a reliable cue. Our results suggest that orientation disparity is an efficient source of information about 3D orientation and that it is plausible that the visual

system could have mechanisms that are sensitive to it. Although orientation disparity is neither necessary nor sufficient for estimating slant, it appears that it could be useful when combined with estimates from position disparity gradients and monocular perspective cues.

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91. A Model for Learning Topographically Organized Parts-Based Representations of Objects in Visual Cortex: Topographic Nonnegative Matrix Factorization

Kenji Hosoda, Masataka Watanabe, Heiko Wersing, Edgar Körner, Hiroshi Tsujino, Hiroshi Tamura, Ichiro Fujita

https://doi.org/10.1162/neco.2009.03-08-722 | pp. 2605-2633

Object representation in the inferior temporal cortex (IT), an area of visual cortex critical for object recognition in the primate, exhibits two prominent properties: (1) objects are represented by the combined activity of columnar clusters of neurons, with each cluster representing component features or parts of objects, and (2) closely related features are continuously represented along the tangential direction of individual columnar clusters. Here we propose a learning model that reflects these properties of parts-based representation and topographic organization in a unified framework. This model is based on a nonnegative matrix factorization (NMF) basis decomposition method. NMF alone provides a parts-based representation where nonnegative inputs are approximated by additive combinations of nonnegative basis functions. Our proposed model of topographic NMF (TNMF) incorporates neighborhood connections between NMF basis functions arranged on a topographic map and attains the topographic property without losing the parts-based property of the NMF. The TNMF represents an input by multiple activity peaks to describe diverse information, whereas conventional topographic models, such as the self-organizing map (SOM), represent an input by a single activity peak in a topographic map. We demonstrate the parts-based and topographic properties of the TNMF by constructing a hierarchical model for object recognition where the TNMF is at the top tier for learning high-level object features. The TNMF showed better generalization performance over NMF for a data set of continuous view change of an image and more robustly preserving the continuity of the view change in its object representation. Comparison of the outputs of our model with actual neural responses recorded in the IT indicates that the TNMF reconstructs the neuronal responses better than the SOM, giving plausibility to the parts-based learning of the model.

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92. Evaluation of Trajectory Planning Models for Arm-Reaching Movements Based on Energy Cost

Jun Nishii, Yoshiaki Taniai

https://doi.org/10.1162/neco.2009.06-08-798 | pp. 2634-2647

Computational studies have suggested that many characteristics of reaching trajectories in a horizontal plane can be effectively predicted by certain models, including, the minimum end point variance model and minimum torque change model. It has also been reported that these characteristics appear to differ from those obtained by the minimum energy cost model that has been reported to explain the characteristics of locomotor patterns. Do these results imply that the human nervous system uses different strategies to resolve the redundancy problem for different tasks? In order to reexamine the optimality of reaching trajectories from a viewpoint of energy cost, we considered the corrective submovements to compensate for

positional error due to signal-dependent noise in motor commands and computed the expected value of the total energy costs required to reach a target by repetition of submovements planned by each of the following models: the minimum energy cost model, minimum end point variance model, and minimum torque change model. The results revealed that when the noise is large, the total energy cost required by the minimum end point variance model and the minimum torque change model can be lower than that required by the minimum energy cost model which assumes minimizing energy cost under noise-free condition. This result indicates that the minimization of the expected value of the energy cost would be an important factor in determining the reaching trajectories.

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93. Filtering Out Deep Brain Stimulation Artifacts Using a Nonlinear Oscillatory Model

Tetiana Aksenova, Dimitri Nowicki, Alim■Louis Benabid

https://doi.org/10.1162/neco.2009.01-07-441 | pp. 2648-2666

This letter is devoted to the suppression of spurious signals (artifacts) in records of neural activity during deep brain stimulation. An approach based on nonlinear adaptive model with self-oscillations is proposed. We developed an algorithm of adaptive filtering based on this approach. The proposed algorithm was tested using recordings collected from patients during the stimulation. This was then compared to existing methods and showed the best performance.

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94. Limited Stochastic Meta-Descent for Kernel-Based Online Learning

Wenwu He

https://doi.org/10.1162/neco.2009.07-08-809 | pp. 2667-2686

To improve the single-run performance of online learning and reinforce its stability, we consider online learning with limited adaptive learning rate in this letter. The letter extends convergence proofs for NORMA to a range of step sizes, then employs support vector learning with stochastic meta-descent (SVMD) limited to that range for step size adaptation, so as to obtain an online kernel algorithm that combines theoretical convergence guarantees with good practical performance. Experiments on different data sets corroborate theoretical results well and show that our method is another promising way for online learning.

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95. A Bound on Modeling Error in Observable Operator Models and an Associated Learning Algorithm

Mingjie Zhao, Herbert Jaeger, Michael R. Thon

https://doi.org/10.1162/neco.2009.01-08-687 | pp. 2687-2712

Observable operator models (OOMs) generalize hidden Markov models (HMMs) and can be represented in a structurally similar matrix formalism. The mathematical theory of OOMs gives rise to a family of constructive, fast, and asymptotically correct learning algorithms, whose statistical efficiency, however, depends crucially on the optimization of two auxiliary transformation matrices. This optimization task is nontrivial; indeed, even formulating computationally accessible optimality criteria is not easy. Here we derive how a bound on the modeling error of an OOM can be expressed in terms of these auxiliary matrices, which

in turn yields an optimization procedure for them and finally affords us with a complete learning algorithm: the error-controlling algorithm. Models learned by this algorithm have an assured error bound on their parameters. The performance of this algorithm is illuminated by comparisons with two types of HMMs trained by the expectation-maximization algorithm, with the efficiency-sharpening algorithm, another recently found learning algorithm for OOMs, and with predictive state representations (Littman & Sutton, 2001) trained by methods representing the state of the art in that field.

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96. Erratum ("Fast and Robust Learning by Reinforcement Signals: Explorations in the Insect Brain" by Ramón Huerta and Thomas Nowotny, Neural Computation, August 2009, Vol. 21, No. 8: 2123–2151)

(authors unknown)

https://doi.org/10.1162/neco.2009.21.9.2713 | pp. 2713-2713

In the August 2009 issue of Neural Computation (Volume 21, Number 8), the letter "Fast and Robust Learning by Reinforcement Signals: Explorations in the Insect Brain" by Ramón Huerta and Thomas Nowotny contained an error in the caption of Figure 11 (p. 2143). The symbols described in the caption were transposed.The printed version reads:Figure 11: Comparison of an SVM trained on partial data sets (squares) and the insect brain after limited number of seen digits (triangles).The corrected version should read:Figure 11: Comparison of an SVM trained on partial data sets (triangles) and the insect brain after limited number of seen digits (squares).

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97. Experience-Induced Neural Circuits That Achieve High Capacity

Vitaly Feldman, Leslie G. Valiant

https://doi.org/10.1162/neco.2009.08-08-851 | pp. 2715-2754

Over a lifetime, cortex performs a vast number of different cognitive actions, mostly dependent on experience. Previously it has not been known how such capabilities can be reconciled, even in principle, with the known resource constraints on cortex, such as low connectivity and low average synaptic strength. Here we describe neural circuits and associated algorithms that respect the brain's most basic resource constraints and support the execution of high numbers of cognitive actions when presented with natural inputs. Our circuits simultaneously support a suite of four basic kinds of task, each requiring some circuit modification: hierarchical memory formation, pairwise association, supervised memorization, and inductive learning of threshold functions. The capacity of our circuits is established by experiments in which sequences of several thousand such actions are simulated by computer and the circuits created tested for subsequent efficacy. Our underlying theory is apparently the only biologically plausible systems-level theory of learning and memory in cortex for which such a demonstration has been performed, and we argue that no general theory of information processing in the brain can be considered viable without such a demonstration.

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98. Operant Matching as a Nash Equilibrium of an Intertemporal Game

Yonatan Loewenstein, Dražen Prelec, H. Sebastian Seung

https://doi.org/10.1162/neco.2009.09-08-854 | pp. 2755-2773

Over the past several decades, economists, psychologists, and neuroscientists have conducted experiments in which a subject, human or animal, repeatedly chooses between alternative actions and is rewarded based on choice history. While individual choices are unpredictable, aggregate behavior typically follows Herrnstein's matching law: the average reward per choice is equal for all chosen alternatives. In general, matching behavior does not maximize the overall reward delivered to the subject, and therefore matching appears inconsistent with the principle of utility maximization. Here we show that matching can be made consistent with maximization by regarding the choices of a single subject as being made by a sequence of multiple selves-one for each instant of time. If each self is blind to the state of the world and discounts future rewards completely, then the resulting game has at least one Nash equilibrium that satisfies both Herrnstein's matching law and the unpredictability of individual choices. This equilibrium is, in general, Pareto suboptimal, and can be understood as a mutual defection of the multiple selves in an intertemporal prisoner's dilemma. The mathematical assumptions about the multiple selves should not be interpreted literally as psychological assumptions. Human and animals do remember past choices and care about future rewards. However, they may be unable to comprehend or take into account the relationship between past and future. This can be made more explicit when a mechanism that converges on the equilibrium, such as reinforcement learning, is considered. Using specific examples, we show that there exist behaviors that satisfy the matching law but are not Nash equilibria. We expect that these behaviors will not be observed experimentally in animals and humans. If this is the case, the Nash equilibrium formulation can be regarded as a refinement of Herrnstein's matching law.

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99. Stimulus-Dependent Correlations and Population Codes

Krešimir Josi■■, Eric Shea■Brown, Brent Doiron, Jaime de la Rocha

https://doi.org/10.1162/neco.2009.10-08-879 | pp. 2774-2804

The magnitude of correlations between stimulus-driven responses of pairs of neurons can itself be stimulus dependent. We examine how this dependence affects the information carried by neural populations about the stimuli that drive them. Stimulus-dependent changes in correlations can both carry information directly and modulate the information separately carried by the firing rates and variances. We use Fisher information to quantify these effects and show that, although stimulus-dependent correlations often carry little information directly, their modulatory effects on the overall information can be large. In particular, if the stimulus dependence is such that correlations increase with stimulus-induced firing rates, this can significantly enhance the information of the population when the structure of correlations is determined solely by the stimulus. However, in the presence of additional strong spatial decay of correlations, such stimulus dependence may have a negative impact. Opposite relationships hold when correlations decrease with firing rates.

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100. Receptive Field Self-Organization in a Model of the Fine Structure in V1 Cortical Columns

Jörg Lücke

https://doi.org/10.1162/neco.2009.07-07-584 | pp. 2805-2845

We study a dynamical model of processing and learning in the visual cortex, which reflects the anatomy of V1 cortical columns and properties of their neuronal receptive fields. Based on recent results on the

fine-scale structure of columns in V1, we model the activity dynamics in subpopulations of excitatory neurons and their interaction with systems of inhibitory neurons. We find that a dynamical model based on these aspects of columnar anatomy can give rise to specific types of computations that result in self-organization of afferents to the column. For a given type of input, self-organization reliably extracts the basic input components represented by neuronal receptive fields. Self-organization is very noise tolerant and can robustly be applied to different types of input. To quantitatively analyze the system's component extraction capabilities, we use two standard benchmarks: the bars test and natural images. In the bars test, the system shows the highest noise robustness reported so far. If natural image patches are used as input, self-organization results in Gabor-like receptive fields. In quantitative comparison with in vivo measurements, we find that the obtained receptive fields capture statistical properties of V1 simple cells that algorithms such as independent component analysis or sparse coding do not reproduce.

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101. A Dynamic Causal Model of the Coupling Between Pulse Stimulation and Neural Activity

Véronique Lefebvre, Ying Zheng, Chris Martin, Ian M. Devonshire, Sam Harris, John Mayhew
https://doi.org/10.1162/neco.2009.07-08-820 | pp. 2846-2868

We present a dynamic causal model that can explain context-dependent changes in neural responses, in the rat barrel cortex, to an electrical whisker stimulation at different frequencies. Neural responses were measured in terms of local field potentials. These were converted into current source density (CSD) data, and the time series of the CSD sink was extracted to provide a time series response train. The model structure consists of three layers (approximating the responses from the brain stem to the thalamus and then the barrel cortex), and the latter two layers contain nonlinearly coupled modules of linear second-order dynamic systems. The interaction of these modules forms a nonlinear regulatory system that determines the temporal structure of the neural response amplitude for the thalamic and cortical layers. The model is based on the measured population dynamics of neurons rather than the dynamics of a single neuron and was evaluated against CSD data from experiments with varying stimulation frequency (1-40 Hz), random pulse trains, and awake and anesthetized animals. The model parameters obtained by optimization for different physiological conditions (anesthetized or awake) were significantly different. Following Friston, Mechelli, Turner, and Price (2000), this work is part of a formal mathematical system currently being developed (Zheng et al., 2005) that links stimulation to the blood oxygen level dependent (BOLD) functional magnetic resonance imaging (fMRI) signal through neural activity and hemodynamic variables. The importance of the model described here is that it can be used to invert the hemodynamic measurements of changes in blood flow to estimate the underlying neural activity.

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102. A Neurocomputational Model for Cocaine Addiction

Amir Dezfouli, Payam Piray, Mohammad Mahdi Keramati, Hamed Ekhtiari, Caro Lucas, Azarakhsh Mokri
https://doi.org/10.1162/neco.2009.10-08-882 | pp. 2869-2893

Based on the dopamine hypotheses of cocaine addiction and the assumption of decrement of brain reward system sensitivity after long-term drug exposure, we propose a computational model for cocaine addiction. Utilizing average reward temporal difference reinforcement learning, we incorporate the elevation of basal reward threshold after long-term drug exposure into the model of drug addiction proposed by Redish. Our

model is consistent with the animal models of drug seeking under punishment. In the case of nondrug reward, the model explains increased impulsivity after long-term drug exposure. Furthermore, the existence of a blocking effect for cocaine is predicted by our model.

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103. Sequential Monte Carlo Point-Process Estimation of Kinematics from Neural Spiking Activity for Brain-Machine Interfaces

Yiwen Wang, António R. C. Paiva, José C. Pr■■ncipe, Justin C. Sanchez

https://doi.org/10.1162/neco.2009.01-08-699 | pp. 2894-2930

Many decoding algorithms for brain machine interfaces' (BMIs) estimate hand movement from binned spike rates, which do not fully exploit the resolution contained in spike timing and may exclude rich neural dynamics from the modeling. More recently, an adaptive filtering method based on a Bayesian approach to reconstruct the neural state from the observed spike times has been proposed. However, it assumes and propagates a gaussian distributed state posterior density, which in general is too restrictive. We have also proposed a sequential Monte Carlo estimation methodology to reconstruct the kinematic states directly from the multichannel spike trains. This letter presents a systematic testing of this algorithm in a simulated neural spike train decoding experiment and then in BMI data. Compared to a point-process adaptive filtering algorithm with a linear observation model and a gaussian approximation (the counterpart for point processes of the Kalman filter), our sequential Monte Carlo estimation methodology exploits a detailed encoding model (tuning function) derived for each neuron from training data. However, this added complexity is translated into higher performance with real data. To deal with the intrinsic spike randomness in online modeling, several synthetic spike trains are generated from the intensity function estimated from the neurons and utilized as extra model inputs in an attempt to decrease the variance in the kinematic predictions. The performance of the sequential Monte Carlo estimation methodology augmented with this synthetic spike input provides improved reconstruction, which raises interesting questions and helps explain the overall modeling requirements better.

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104. Correlation Between Eigenvalue Spectra and Dynamics of Neural Networks

Qingguo Zhou, Jin Tao, Hong Zhao

https://doi.org/10.1162/neco.2009.12-07-671 | pp. 2931-2941

This letter presents a study of the correlation between the eigenvalue spectra of synaptic matrices and the dynamical properties of asymmetric neural networks with associative memories. For this type of neural network, it was found that there are essentially two different dynamical phases: the chaos phase, with almost all trajectories converging to a single chaotic attractor, and the memory phase, with almost all trajectories being attracted toward fixed-point attractors acting as memories. We found that if a neural network is designed in the chaos phase, the eigenvalue spectrum of its synaptic matrix behaves like that of a random matrix (i.e., all eigenvalues lie uniformly distributed within a circle in the complex plan), and if it is designed in the memory phase, the eigenvalue spectrum will split into two parts: one part corresponds to a random background, the other part equal in number to the memory attractors. The mechanism for these phenomena is discussed in this letter.

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105. Distance Learning in Discriminative Vector Quantization

Petra Schneider, Michael Biehl, Barbara Hammer

https://doi.org/10.1162/neco.2009.10-08-892 | pp. 2942-2969

Discriminative vector quantization schemes such as learning vector quantization (LVQ) and extensions thereof offer efficient and intuitive classifiers based on the representation of classes by prototypes. The original methods, however, rely on the Euclidean distance corresponding to the assumption that the data can be represented by isotropic clusters. For this reason, extensions of the methods to more general metric structures have been proposed, such as relevance adaptation in generalized LVQ (GLVQ) and matrix learning in GLVQ. In these approaches, metric parameters are learned based on the given classification task such that a data-driven distance measure is found. In this letter, we consider full matrix adaptation in advanced LVQ schemes. In particular, we introduce matrix learning to a recent statistical formalization of LVQ, robust soft LVQ, and we compare the results on several artificial and real-life data sets to matrix learning in GLVQ, a derivation of LVQ-like learning based on a (heuristic) cost function. In all cases, matrix adaptation allows a significant improvement of the classification accuracy. Interestingly, however, the principled behavior of the models with respect to prototype locations and extracted matrix dimensions shows several characteristic differences depending on the data sets.

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106. An Integral Upper Bound for Neural Network Approximation

Paul C. Kainen, V■ra K■rková

https://doi.org/10.1162/neco.2009.04-08-745 | pp. 2970-2989

Complexity of one-hidden-layer networks is studied using tools from nonlinear approximation and integration theory. For functions with suitable integral representations in the form of networks with infinitely many hidden units, upper bounds are derived on the speed of decrease of approximation error as the number of network units increases. These bounds are obtained for various norms using the framework of Bochner integration. Results are applied to perceptron networks.

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107. Erratum (“Learning with ‘Relevance’: Using a Third Factor to Stabilize Hebbian Learning” by Bernd Porr and Florentin Wörgötter, Neural Computation, October 2007, Vol. 19, No. 10: 2694–2719)

(authors unknown)

https://doi.org/10.1162/neco.2009.21.10.2990b | pp. 2990-2990

No abstract available.

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108. Erratum (“On the Asymptotic Equivalence Between Differential Hebbian and Temporal Difference Learning” by Christoph Kolodziejski, Bernd Porr, and Florentin Wörgötter, Neural Computation, April 2009, Vol. 21, No. 4: 1173–1202)

(authors unknown)

https://doi.org/10.1162/neco.2009.21.10.2990c | pp. 2990-2990

No abstract available.

109. Erratum (“Strongly Improved Stability and Faster Convergence of Temporal Sequence Learning by Using Input Correlations Only” by Bernd Porr and Florentin Wörgötter, Neural Computation, June 2006, Vol. 18, No. 6: 1380–1412)

(authors unknown)

https://doi.org/10.1162/neco.2009.21.10.2990a | pp. 2990-2990

No abstract available.

110. On the Maximization of Information Flow Between Spiking Neurons

Lucas C. Parra, Jeffrey M. Beck, Anthony J. Bell

https://doi.org/10.1162/neco.2009.04-06-184 | pp. 2991-3009

A feedforward spiking network represents a nonlinear transformation that maps a set of input spikes to a set of output spikes. This mapping transforms the joint probability distribution of incoming spikes into a joint distribution of output spikes. We present an algorithm for synaptic adaptation that aims to maximize the entropy of this output distribution, thereby creating a model for the joint distribution of the incoming point processes. The learning rule that is derived depends on the precise pre- and postsynaptic spike timings. When trained on correlated spike trains, the network learns to extract independent spike trains, thereby uncovering the underlying statistical structure and creating a more efficient representation of the incoming spike trains.

111. Cortical Circuitry Implementing Graphical Models

Shai Litvak, Shimon Ullman

https://doi.org/10.1162/neco.2009.05-08-783 | pp. 3010-3056

In this letter, we develop and simulate a large-scale network of spiking neurons that approximates the inference computations performed by graphical models. Unlike previous related schemes, which used sum and product operations in either the log or linear domains, the current model uses an inference scheme based on the sum and maximization operations in the log domain. Simulations show that using these operations, a large-scale circuit, which combines populations of spiking neurons as basic building blocks, is capable of finding close approximations to the full mathematical computations performed by graphical models within a few hundred milliseconds. The circuit is general in the sense that it can be wired for any graph structure, it supports multistate variables, and it uses standard leaky integrate-and-fire neuronal units. Following previous work, which proposed relations between graphical models and the large-scale cortical anatomy, we focus on the cortical microcircuitry and propose how anatomical and physiological aspects of the local circuitry may map onto elements of the graphical model implementation. We discuss in particular the roles of three major types of inhibitory neurons (small fast-spiking basket cells, large layer 2/3 basket

cells, and double-bouquet neurons), subpopulations of strongly interconnected neurons with their unique connectivity patterns in different cortical layers, and the possible role of minicolumns in the realization of the population-based maximum operation.

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112. Electrical Coupling Promotes Fidelity of Responses in the Networks of Model Neurons

Georgi S. Medvedev

https://doi.org/10.1162/neco.2009.07-08-813 | pp. 3057-3078

We consider an integrate-and-fire element subject to randomly perturbed synaptic input and an electrically coupled ensemble of such elements. The latter is interpreted as either a model of electrically coupled population of neurons or a multicompartment model of a dendrite. Random fluctuations blur the input signal and cause false responses in the system dynamics. For instance, under the influence of noise, the system may respond with an action potential to a subthreshold stimulus. We show that the responses of the elements within the network are more reliable than the responses of the same elements in isolation. Specifically, we show that the variances of the stochastic processes generated by the coupled model can be made arbitrarily small (i.e., the network responses can be made arbitrarily accurate) by increasing the number of elements in the network and the strength of electrical coupling. Our results suggest that the organization of cells in electrically coupled groups on the network level, or the dendritic morphology on the cellular level, may be involved in the filtering noise and therefore may play an important role in the information processing mechanisms operating on the network or cellular level respectively.

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113. Maximum Likelihood Decoding of Neuronal Inputs from an Interspike Interval Distribution

Xuejuan Zhang, Gongqiang You, Tianping Chen, Jianfeng Feng

https://doi.org/10.1162/neco.2009.06-08-807 | pp. 3079-3105

An expression for the probability distribution of the interspike interval of a leaky integrate-and-fire (LIF) model neuron is rigorously derived, based on recent theoretical developments in the theory of stochastic processes. This enables us to find for the first time a way of developing maximum likelihood estimates (MLE) of the input information (e.g., afferent rate and variance) for an LIF neuron from a set of recorded spike trains. Dynamic inputs to pools of LIF neurons both with and without interactions are efficiently and reliably decoded by applying the MLE, even within time windows as short as 25 msec.

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114. Classification of Correlated Patterns with a Configurable Analog VLSI Neural Network of Spiking Neurons and Self-Regulating Plastic Synapses

M. Giulioni, Mario Pannunzi, D. Badoni, Vittorio Dante, Paolo Del Giudice

https://doi.org/10.1162/neco.2009.08-07-599 | pp. 3106-3129

We describe the implementation and illustrate the learning performance of an analog VLSI network of 32 integrate-and-fire neurons with spike-frequency adaptation and 2016 Hebbian bistable spike-driven stochastic synapses, endowed with a self-regulating plasticity mechanism, which avoids unnecessary

synaptic changes. The synaptic matrix can be flexibly configured and provides both recurrent and external connectivity with address-event representation compliant devices. We demonstrate a marked improvement in the efficiency of the network in classifying correlated patterns, owing to the self-regulating mechanism.

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115. Approximate Learning Algorithm in Boltzmann Machines

Muneki Yasuda, Kazuyuki Tanaka

https://doi.org/10.1162/neco.2009.08-08-844 | pp. 3130-3178

Boltzmann machines can be regarded as Markov random fields. For binary cases, they are equivalent to the Ising spin model in statistical mechanics. Learning systems in Boltzmann machines are one of the NP-hard problems. Thus, in general we have to use approximate methods to construct practical learning algorithms in this context. In this letter, we propose new and practical learning algorithms for Boltzmann machines by using the belief propagation algorithm and the linear response approximation, which are often referred as advanced mean field methods. Finally, we show the validity of our algorithm using numerical experiments.

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116. Robust Kernel Principal Component Analysis

Su-Yun Huang, Yi-Ren Yeh, Shinto Eguchi

https://doi.org/10.1162/neco.2009.02-08-706 | pp. 3179-3213

This letter discusses the robustness issue of kernel principal component analysis. A class of new robust procedures is proposed based on eigenvalue decomposition of weighted covariance. The proposed procedures will place less weight on deviant patterns and thus be more resistant to data contamination and model deviation. Theoretical influence functions are derived, and numerical examples are presented as well. Both theoretical and numerical results indicate that the proposed robust method outperforms the conventional approach in the sense of being less sensitive to outliers. Our robust method and results also apply to functional principal component analysis.

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117. Adaptive Capability of Recurrent Neural Networks with Fixed Weights for Series-Parallel System Identification

James Lo

https://doi.org/10.1162/neco.2009.06-07-542 | pp. 3214-3227

By a fundamental neural filtering theorem, a recurrent neural network with fixed weights is known to be capable of adapting to an uncertain environment. This letter reports some mathematical results on the performance of such adaptation for series-parallel identification of a dynamical system as compared with the performance of the best series-parallel identifier possible under the assumption that the precise value of the uncertain environmental process is given. In short, if an uncertain environmental process is observable (not necessarily constant) from the output of a dynamical system or constant (not necessarily observable), then a recurrent neural network exists as a series-parallel identifier of the dynamical system whose output approaches the output of an optimal series-parallel identifier using the environmental process as an additional input.

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118. Partial Orders of Similarity Differences Invariant Between EEG-Recorded Brain and Perceptual Representations of Language

Patrick Suppes, Marcos Perreau-Guimaraes, Dik Kin Wong

https://doi.org/10.1162/neco.2009.04-08-764 | pp. 3228-3269

The idea of a hierarchical structure of language constituents of phonemes, syllables, words, and sentences is robust and widely accepted. Empirical similarity differences at every level of this hierarchy have been analyzed in the form of confusion matrices for many years. By normalizing such data so that differences are represented by conditional probabilities, semiorders of similarity differences can be constructed. The intersection of two such orderings is an invariant partial ordering with respect to the two given orders. These invariant partial orderings, especially between perceptual and brain representations, but also for comparison of brain images of words generated by auditory or visual presentations, are the focus of this letter. Data from four experiments are analyzed, with some success in finding conceptually significant invariants.

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119. Is the Homunculus “Aware” of Sensory Adaptation?

Peggy Seriès, Alan A. Stocker, Eero P. Simoncelli

https://doi.org/10.1162/neco.2009.09-08-869 | pp. 3271-3304

Neural activity and perception are both affected by sensory history. The work presented here explores the relationship between the physiological effects of adaptation and their perceptual consequences. Perception is modeled as arising from an encoder-decoder cascade, in which the encoder is defined by the probabilistic response of a population of neurons, and the decoder transforms this population activity into a perceptual estimate. Adaptation is assumed to produce changes in the encoder, and we examine the conditions under which the decoder behavior is consistent with observed perceptual effects in terms of both bias and discriminability. We show that for all decoders, discriminability is bounded from below by the inverse Fisher information. Estimation bias, on the other hand, can arise for a variety of different reasons and can range from zero to substantial. We specifically examine biases that arise when the decoder is fixed, "unaware" of the changes in the encoding population (as opposed to "aware" of the adaptation and changing accordingly). We simulate the effects of adaptation on two well-studied sensory attributes, motion direction and contrast, assuming a gain change description of encoder adaptation. Although we cannot uniquely constrain the source of decoder bias, we find for both motion and contrast that an "unaware" decoder that maximizes the likelihood of the percept given by the preadaptation encoder leads to predictions that are consistent with behavioral data. This model implies that adaptation-induced biases arise as a result of temporary suboptimality of the decoder.

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120. Decoding Movement Trajectories Through a T-Maze Using Point Process Filters Applied to Place Field Data from Rat Hippocampal Region CA1

Yifei Huang, Mark P. Brandon, Amy L. Griffin, Michael E. Hasselmo, Uri T. Eden

https://doi.org/10.1162/neco.2009.10-08-893 | pp. 3305-3334

Firing activity from neural ensembles in rat hippocampus has been previously used to determine an animal's position in an open environment and separately to predict future behavioral decisions. However, a unified statistical procedure to combine information about position and behavior in environments with complex

topological features from ensemble hippocampal activity has yet to be described. Here we present a two-stage computational framework that uses point process filters to simultaneously estimate the animal's location and predict future behavior from ensemble neural spiking activity. First, in the encoding stage, we linearized a two-dimensional T-maze, and used spline-based generalized linear models to characterize the place-field structure of different neurons. All of these neurons displayed highly specific position-dependent firing, which frequently had several peaks at multiple locations along the maze. When the rat was at the stem of the T-maze, the firing activity of several of these neurons also varied significantly as a function of the direction it would turn at the decision point, as detected by ANOVA. Second, in the decoding stage, we developed a state-space model for the animal's movement along a T-maze and used point process filters to accurately reconstruct both the location of the animal and the probability of the next decision. The filter yielded exact full posterior densities that were highly nongaussian and often multimodal. Our computational framework provides a reliable approach for characterizing and extracting information from ensembles of neurons with spatially specific context or task-dependent firing activity.

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121. Selective Population Rate Coding: A Possible Computational Role of Gamma Oscillations in Selective Attention

Naoki Masuda

https://doi.org/10.1162/neco.2009.09-08-857 | pp. 3335-3362

Selective attention is often accompanied by gamma oscillations in local field potentials and spike field coherence in brain areas related to visual, motor, and cognitive information processing. Gamma oscillations are implicated to play an important role in, for example, visual tasks including object search, shape perception, and speed detection. However, the mechanism by which gamma oscillations enhance cognitive and behavioral performance of attentive subjects is still elusive. Using feedforward fan-in networks composed of spiking neurons, we examine a possible role for gamma oscillations in selective attention and population rate coding of external stimuli. We implement the concept proposed by Fries (2005) that under dynamic stimuli, neural populations effectively communicate with each other only when there is a good phase relationship among associated gamma oscillations. We show that the downstream neural population selects a specific dynamic stimulus received by an upstream population and represents it by population rate coding. The encoded stimulus is the one for which gamma rhythm in the corresponding upstream population is resonant with the downstream gamma rhythm. The proposed role for gamma oscillations in stimulus selection is to enable top-down control, a neural version of time division multiple access used in communication engineering.

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122. Taming Fluctuations in a Stochastic Model of Spike-Timing-Dependent Plasticity

Terry Elliott, Konstantinos Lagogiannis

https://doi.org/10.1162/neco.2009.12-08-916 | pp. 3363-3407

A stochastic model of spike-timing-dependent plasticity proposes that single synapses express fixed-amplitude jumps in strength, the amplitudes being independent of the spike time difference. However, the probability that a jump in strength occurs does depend on spike timing. Although the model has a number of desirable features, the stochasticity of response of a synapse introduces potentially large

fluctuations into changes in synaptic strength. These can destabilize the segregated patterns of afferent connectivity characteristic of neuronal development. Previously we have taken these jumps to be small relative to overall synaptic strengths to control fluctuations, but doing so increases developmental timescales unacceptably. Here, we explore three alternative ways of taming fluctuations. First, a calculation of the variance for the change in synaptic strength shows that the mean change eventually dominates fluctuations, but on timescales that are too long. Second, it is possible that fluctuations in strength may cancel between synapses, but we show that correlations between synapses emasculate the law of large numbers. Finally, by separating plasticity induction and expression, we introduce a temporal window during which induction signals are low-pass-filtered before expression. In this way, fluctuations in strength are tamed, stabilizing segregated states of afferent connectivity.

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123. Learning to Discriminate Through Long-Term Changes of Dynamical Synaptic Transmission

Christian Leibold, M. H. K. Bendels

https://doi.org/10.1162/neco.2009.12-08-929 | pp. 3408-3428

Short-term synaptic plasticity is modulated by long-term synaptic changes. There is, however, no general agreement on the computational role of this interaction. Here, we derive a learning rule for the release probability and the maximal synaptic conductance in a circuit model with combined recurrent and feedforward connections that allows learning to discriminate among natural inputs. Short-term synaptic plasticity thereby provides a nonlinear expansion of the input space of a linear classifier, whereas the random recurrent network serves to decorrelate the expanded input space. Computer simulations reveal that the twofold increase in the number of input dimensions through short-term synaptic plasticity improves the performance of a standard perceptron up to 100%. The distributions of release probabilities and maximal synaptic conductances at the capacity limit strongly depend on the balance between excitation and inhibition. The model also suggests a new computational interpretation of spikes evoked by stimuli outside the classical receptive field. These neuronal activities may reflect decorrelation of the expanded stimulus space by intracortical synaptic connections.

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124. Plasticity-Induced Symmetry Relationships Between Adjacent Self-Organizing Topographic Maps

Jared Sylvester, James A. Reggia

https://doi.org/10.1162/neco.2009.04-08-763 | pp. 3429-3443

In many species, adjacent topographic maps in sensory neocortex are found to be oriented as roughly mirror-image copies of one another. Here we use a computational model to show for the first time that, in principle, adjacent cortical topographic maps that are mirror-image symmetric along two dimensions can arise from activity-dependent changes if the distribution radius of afferents sufficiently exceeds that of horizontal intracortical interactions. We also find that infrequently, other types of intermap symmetry and previously unexpected map relationships (such as interlocking rotation, in which two adjacent maps become intertwined) can occur. These results support the hypothesis that activity-dependent synaptic changes play a more important role in forming the orientations of adjacent cortical maps than is currently recognized.

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125. Asymptotic Behavior of Periodic Cohen-Grossberg Neural Networks with Delays

Wei Lin

https://doi.org/10.1162/neco.2009.08-08-840 | pp. 3444-3459

Without assuming the positivity of the amplification functions, we prove some M-matrix criteria for the R(n)(+)-global asymptotic stability of periodic Cohen-Grossberg neural networks with delays. By an extension of the Lyapunov method, we are able to include neural systems with multiple nonnegative periodic solutions and nonexponential convergence rate in our model and also include the Lotka-Volterra system, an important prototype of competitive neural networks, as a special case. The stability criteria for autonomous systems then follow as a corollary. Two numerical examples are provided to show that the limiting equilibrium or periodic solution need not be positive.

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126. Making the Error-Controlling Algorithm of Observable Operator Models Constructive

Mingjie Zhao, Herbert Jaeger, Michael R. Thon

https://doi.org/10.1162/neco.2009.10-08-878 | pp. 3460-3486

Observable operator models (OOMs) are a class of models for stochastic processes that properly subsumes the class that can be modeled by finite-dimensional hidden Markov models (HMMs). One of the main advantages of OOMs over HMMs is that they admit asymptotically correct learning algorithms. A series of learning algorithms has been developed, with increasing computational and statistical efficiency, whose recent culmination was the error-controlling (EC) algorithm developed by the first author. The EC algorithm is an iterative, asymptotically correct algorithm that yields (and minimizes) an assured upper bound on the modeling error. The run time is faster by at least one order of magnitude than EM-based HMM learning algorithms and yields significantly more accurate models than the latter. Here we present a significant improvement of the EC algorithm: the constructive error-controlling (CEC) algorithm. CEC inherits from EC the main idea of minimizing an upper bound on the modeling error but is constructive where EC needs iterations. As a consequence, we obtain further gains in learning speed without loss in modeling accuracy.

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127. Estimation of Sparse Nonnegative Sources from Noisy Overcomplete Mixtures Using MAP

César F. Caiafa, Andrzej Cichocki

https://doi.org/10.1162/neco.2009.08-08-846 | pp. 3487-3518

In this letter, we propose a new algorithm for estimating sparse nonnegative sources from a set of noisy linear mixtures. In particular, we consider difficult situations with high noise levels and more sources than sensors (underdetermined case). We show that when sources are very sparse in time and overlapped at some locations, they can be recovered even with very low signal-to-noise ratio, and by using many fewer sensors than sources. A theoretical analysis based on Bayesian estimation tools is included showing strong connections with algorithms in related areas of research such as ICA, NMF, FOCUSS, and sparse representation of data with overcomplete dictionaries. Our algorithm uses a Bayesian approach by modeling sparse signals through mixed-state random variables. This new model for priors imposes l(0) norm-based

sparsity. We start our analysis for the case of nonoverlapped sources (1-sparse), which allows us to simplify the search of the posterior maximum avoiding a combinatorial search. General algorithms for overlapped cases, such as 2-sparse and k-sparse sources, are derived by using the algorithm for 1-sparse signals recursively. Additionally, a combination of our MAP algorithm with the NN-KSVD algorithm is proposed for estimating the mixing matrix and the sources simultaneously in a real blind fashion. A complete set of simulation results is included showing the performance of our algorithm.

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128. On Blind Separability Based on the Temporal Predictability Method

Shengli Xie, Guoxu Zhou, Zuyuan Yang, Yuli Fu

https://doi.org/10.1162/neco.2009.10-08-890 | pp. 3519-3531

This letter discusses blind separability based on temporal predictability (Stone, 2001 ; Xie, He, & Fu, 2005). Our results show that the sources are separable using the temporal predictability method if and only if they have different temporal structures (i.e., autocorrelations). Consequently, the applicability and limitations of the temporal predictability method are clarified. In addition, instead of using generalized eigendecomposition, we suggest using joint approximate diagonalization algorithms to improve the robustness of the method. A new criterion is presented to evaluate the separation results. Numerical simulations are performed to demonstrate the validity of the theoretical results.

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129. Adaptive Relevance Matrices in Learning Vector Quantization

Petra Schneider, Michael Biehl, Barbara Hammer

https://doi.org/10.1162/neco.2009.11-08-908 | pp. 3532-3561

We propose a new matrix learning scheme to extend relevance learning vector quantization (RLVQ), an efficient prototype-based classification algorithm, toward a general adaptive metric. By introducing a full matrix of relevance factors in the distance measure, correlations between different features and their importance for the classification scheme can be taken into account and automated, and general metric adaptation takes place during training. In comparison to the weighted Euclidean metric used in RLVQ and its variations, a full matrix is more powerful to represent the internal structure of the data appropriately. Large margin generalization bounds can be transferred to this case, leading to bounds that are independent of the input dimensionality. This also holds for local metrics attached to each prototype, which corresponds to piecewise quadratic decision boundaries. The algorithm is tested in comparison to alternative learning vector quantization schemes using an artificial data set, a benchmark multiclass problem from the UCI repository, and a problem from bioinformatics, the recognition of splice sites for C. elegans.

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130. Volume 21 Index by Author

(authors unknown)

https://doi.org/10.1162/neco.2009.21.12.3562 | pp. 3562-3586

No abstract available.

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131. A Binary Variable Model for Affinity Propagation

Inmar E. Givoni, Brendan J. Frey

https://doi.org/10.1162/neco.2008.05-08-785 | pp. 090202092741069-12

No abstract available.

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132. Dynamical Analysis of Bayesian Inference Models for the Eriksen Task

Yuan Sophie Liu, Angela J. Yu, Philip Holmes

https://doi.org/10.1162/neco.2008.03-07-495 | pp. 090202092741069-34

The Eriksen task is a classical paradigm that explores the effects of competing sensory inputs on response tendencies, and the nature of selective attention in controlling these processes.In this task, conflicting flanker stimuli interfere with the processing of a central target, especially on short reaction-time trials.This task has been modeled by neural networks and more recently by a normative Bayesian account.Here, we analyze the dynamics of the Bayesian models, which are nonlinear, coupled discrete-time dynamical systems, by considering simplified, approximate systems that are linear and decoupled.Analytical solutions of these allow us to describe how posterior probabilities and psychometric functions depend upon model parameters.We compare our results with numerical simulations of the original models and derive fits to experimental data, showing that agreements are rather good.We also investigate continuum limits of these simplified dynamical systems, and demonstrate that Bayesian updating is closely related to a drift-diffusion process, whose implementation in neural network models has been extensively studied.This provides insight on how neural substrates can implement Bayesian computations.

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133. Generation of Spike Trains with Controlled Auto- and Cross-Correlation Functions

Michael Krumin, Shy Shoham

https://doi.org/10.1162/neco.2008.08-08-847 | pp. 090202092741069-23

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134. A New Hypothesis for Sleep: Tuning for Criticality

Barak A. Pearlmutter, Conor Houghton

https://doi.org/10.1162/neco.2008.05-08-787 | pp. 090202092741069-20

We propose that the critical function of sleep is to prevent uncontrolled neuronal feedback while allowing rapid responses and prolonged retention of short-term memories. The goal of learning is optimal behavior, and this sometimes requires the integration of sensory stimuli that are widely separated in time. At a neuronal level, this corresponds to persistent activity in local networks. Unfortunately, when a network exhibits persistent activity, small changes in the parameters or conditions can lead to runaway oscillations. Thus, the very changes that improve the processing performance of the network can put it at risk of runaway oscillation. To prevent this, stimulusdependent plasticity should only be permitted when there is a margin of safety around the current network parameters. We propose that a critical role of sleep is to establish a

margin of safety by exposing the network to a variety of conditions and inputs, observing for erratic behavior, and-adjusting the parameters accordingly. During wakefulness this margin of safety is gradually consumed, ultimately requiring refreshment by another period of sleep. When sleep is not possible, an emergency mechanism comes into play to prevent runaway oscillations; this is done at the expense of processing efficiency, and constitutes tiredness. We contend that this theory matches the phenomenology of sleep and tiredness. It also makes a number of novel testable predictions.

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135. Capacity of a Single Spiking Neuron Channel

Shiro Ikeda, Jonathan H. Manton

https://doi.org/10.1162/neco.2008.05-08-792 | pp. 090202092741069-35

No abstract available.

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136. How Does Maintenance of Network Activity Depend on Endogenous Dynamics of Isolated Neurons?

Andrey V. Olypher, Ronald L. Calabrese

https://doi.org/10.1162/neco.2008.01-08-685 | pp. 090202092741069-18

No abstract available.